

Annual Report 1989-90



**NATIONAL ACADEMY OF
AGRICULTURAL RESEARCH MANAGEMENT**
Rajendranagar, Hyderabad-500 030 (India)

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INTRODUCTION

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1.1 Historical Background

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The concept of management of agricultural research is fast gaining ground in the country. Management principles are being increasingly applied in agricultural research, education and extension. The need for an institutional set-up to provide training on principles of scientific management of agricultural research was felt two decades ago by the Indian Council of Agricultural Research (ICAR) and the Government of India, following the organisation of a training course in Agricultural Project Management, in response to a request from the FAO. A National Training Centre for Agriculture Development and Project Planning functioned at IARI, New Delhi, during the Fourth Five Year Plan. This successful venture further emphasised the need to continue these training activities, and to establish an organisation for this purpose.

A working group, set up during the Fifth Five Year Plan, recommended the establishment of a Central Staff College for Agriculture to function as national and international repository of ideas and information in the field of agricultural research and development. The proposed college was also required to act as a clearing house for dissemination and utilisation of Research Management information and also to undertake a programme of publishing high quality training material and allied publications considered important for the priority areas of education.

After more detailed deliberations and planning, the Central Staff College for Agriculture came into existence on September 1, 1976. This was later renamed as the National Academy of Agricultural Research Management. The Academy is located at Rajendranagar, Hyderabad, at the campus of the Andhra Pradesh Agricultural University (APAU) on 50 hectares of land provided by the University to the Academy on a long-term lease. The NAARM campus is situated at Rajendranagar, a suburb of Hyderabad, about 15 km south of the City

Several functional buildings have now been constructed in this campus. These include an Administrative Block, Faculty-cum-Laboratory Building, an Academic Block which has several class and discussion rooms, Halls of Residence and Scientists Home. Besides, residential quarters for all categories of staff have also been constructed. Several instructional and technology demonstration facilities have been developed. An Advanced Centre on Agricultural Education and research Management has also been established with the support of UNDP/ UNESCO/ FAO. These physical and infrastructural facilities as well as the technical strength of the programmes have placed the Academy as an important National and International Centre serving the country and the South and South East Asian Region.

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OBJECTIVES

The National Academy of Agricultural Research Management (NAARM) was established with the main objective of familiarising all those concerned with the Agricultural Research, Education and Development with the principles of scientific management. Its mandate includes the responsibility to plan, organize and conduct training programmes, workshops and seminars in the various areas related to agricultural research management for scientists of ICAR and agricultural universities. The Academy also organises training programmes in the areas of educational technology, curriculum development and structured learning.

The main objectives of the Academy are:

- * to organise and conduct training programmes in agricultural research management to scientists at various levels,
- * to build up high quality resource materials in the subject of agricultural research management based on actual field experience,
- * to undertake systematic review and study of management problems of ICAR Research institutes, Agricultural universities, Agricultural research projects and systems,
- * to organise, liaise and coordinate programmes of international cooperation in the field of agricultural research management,
- * to plan, organise and conduct training programmes, workshops and seminars in educational technology and structured learning for the faculty of agricultural universities and ICAR institutes, and,
- * to assist the State Agricultural Universities (SAUs) in developing regional centres of management to cater to the needs of the various SAUs

The Academy organises various types of training programmes to fulfil the needs of the clientele groups working at various levels of management in the agricultural research and development organisations. Three levels of clientele -- Junior, Middle and Senior-- are identified for training purposes.

2.1 COURSE FOR JUNIOR SCIENTISTS

Foundation Course on Agricultural Research Project Management (FCARPM).

A regular course offered to the scientists recruited to Agricultural Research Service by the ICAR, it aims at providing exposure to the trainees on the concepts and principles of project management with special emphasis on project formulation and implementation. It also includes instruction in related areas such as group dynamics, interpersonal relationships, interdisciplinary project formulation and motivation management which would shape them as effective research workers. The course is organised within the framework of the following objectives:

- * to acquaint the scientists with the general agricultural scenario in the country with regard to research and development,
- * to present an insight into the rural society and structure,
- * to equip them with the techniques in Research Project Management,
- * to impart skills in organisational behaviour for high motivation, team work and conflict resolution,
- * to orient the scientists to the organisation of the Indian Council of Agricultural Research and the requirements of the Agricultural Research Service, and,
- * to foster a feeling of fraternity among the members of ARS serving in different parts of the country

The training programme lasts for five months. The first phase includes six weeks' training at the Academy, oriented mainly to the basics of Agricultural Research Management. Two months of training, called Field Experience Training(FET) at a regional station of National Agricultural Research Project of

ICAR institute, constitutes the second phase. The FET is designed to impart training in project identification and formulation. This phase helps the trainees develop an appreciation for rural life and environment. In the third phase, comprising six weeks of training at the Academy, the trainees are given exposure in areas which would strengthen their skills in agricultural research management, scientific writing and presentation, and improvement of communication skills.

During the period under report, 151 ARS scientists, freshly recruited into the Service, completed their training. Currently, 47 ARS scientists, are receiving training. The following course contents are covered in the training programme:

I. General Agricultural Scenario in the Country

1. NAARM and its Role
2. Agricultural Research Management Process
3. Agroecological Zones and Cropping Systems in the World
4. Agricultural Production---Potential and Constraints
5. Livestock Production---Potential and Constraints
6. Impact of Green Revolution---Strategies for Future

II. Rural Society and Structure

1. Rural Development ; Basic Issues and Suggested Policy Frame
2. Rural Leadership and Rural Development Programmes
3. Socio-economic Surveys and Tools of Data Collection for Socio-economic surveys
4. Analysis of Data of Socio-economic Survey for Formulating and Preparing Viable Research Projects

III. Research Project

1. Creativity and Problem Solving
2. Research Programme Planning
3. The Research Project
4. Project Appraisal
5. Project Selection Exercise
6. Project Management Techniques--CPM & PERT
7. Resource Smoothing
8. Uncertainties in Research

IV. Transfer and Adoption of Technology

1. Technology Transfer and Appropriate Technology
2. Institutional Structures for Transfer of Technology
3. Treatment of Research Findings
4. Effective Presentation Techniques in Scientific Communication
5. Communication through A.V.Aids
6. Communication through Mass Media
7. Development Communication in Agriculture
8. Constraint Analysis in Research & Adoption
9. Diffusion Process in Agriculture

V. History, Structure and organisation of Agricultural Research

1. History and Growth of ICAR
2. Structure and Organisation of ICAR
3. Agricultural Research Service
4. National Agricultural Research Project
5. Farming Systems Research
6. National Agricultural Research System in India
7. Systems Approach for Technology Development
8. Brief Review of Agricultural Research Systems in the World
9. Linkages and Coordination between ICAR and Agricultural Universities-Role of ICAR
10. Linkages of ICAR with Other National Research Organisations
11. Research and Extension Systems in Agricultural Universities
12. All India Coordinated Research Projects of ICAR

VI. Organisational Behaviour

1. Interpersonal Relationships
2. Stereotypes

VII. Team Work and Interpersonal Relationships

1. Individual and the Organisation
2. Groups in Organisation
3. Team Work
4. Group Dynamics--Exercise
5. Motivation and Behaviour
6. Motivation and Work
7. Exercise on Motivation

8. Transactional Analysis
9. Conflicts in Organisations
10. Positive Thinking

VIII. Training in Skills Development

1. Communication Skills
2. Rapid Reading and Comprehension
3. Basics of Computers and Use of Computers in Agricultural Research and Education
4. Statistical Inferences
5. Designs of Experiments
6. Basic Concepts of Linear Programming
7. Proof Reading
8. Production and Printing Process

IX. Economics of Agricultural Development

1. Agricultural and Economic Development with reference to Five Year Plans
2. Agricultural Labour and Employment
3. Agricultural Finance
4. Agricultural Prices
5. Land Systems and Measures of Reform and its impact on Agricultural Productivity

X. Organisation of Research

1. Internal Research Organisations-ICAR and International collaboration in Research
2. Research Projects Monitoring and evaluation in ICAR
3. Contribution of ICAR to Agricultural Education Systems in the Country
4. Consequences Technologies in Agriculture

XI. Information Management

1. Sources of Agricultural Information
2. Citation Indexing
3. Management of Personal Documents
4. Book Reviewing in the Sciences and Physical parts of a book
5. Poster Presentation
6. Writing of Scientific Paper
7. Technical Report Writing

8. Writing of Articles for Green and Glory
9. Review of Literature
10. Transfer of scientific information
11. On-line Information Services and Electronic Databases

XII. Recent trends in Research

1. Crop Sciences
2. Oil Seeds
3. Dryland Agriculture
4. Pest Management
5. Soil Management
6. Fisheries Management
7. Livestock Management
8. Horticultural Crop
9. Poultry Management
10. Farm Machinery and Power
11. Irrigation and Water Management
12. Energy Management in Agriculture
13. Solar and Wind Energy
14. Biomass and Energy
15. Pollution

XIII. Administration and Accounts

1. General Service Conditions -An Overview
2. Stores Management and Purchase Procedures
3. Project Budgeting and Monitoring
4. Management of Research Stations

XIV. PRACTICALS I

1. Audio -Visual aids
2. Treatment of research findings
3. Library

XV. PRACTICALS II

1. Green Revolution Game
2. Book Review presentations
3. Field visit for Field Experience Training

XVI. Visits

1. ICRISAT
2. National Institute of Nutrition
3. National Remote Sensing Agency
4. Other ICAR institutes

XVII. Others

1. Group seminar presentations
2. Feedback on group seminar presentations
3. Socio-cultural field work

2.2 COURSES FOR MIDDLE AND SENIOR SCIENTISTS

Course on Agricultural Research Project Management (CARPM)

The course is designed to provide exposure to the research project managers in the ICAR institutes to the concepts of research projects and application of 'management approach' while dealing with the problems faced by the agricultural research units. It is offered to the senior research managers heading divisions in the ICAR institutes and Professors and Heads of Departments in State Agricultural Universities. The duration of the course is two weeks. This course was offered twice since during this period ---from June 13 to 24, 1989 and November 21 to Dec 2 1989.

Twenty eight participants attended the course.

The topics covered in the course include:

Project Planning --Importance of Planning Long-term and short-term projects in the management process and systematic project planning.

Project selection -Methods and criteria used for project selection.

Monitoring and Evaluation --Network methods for time scheduling, control and monitoring and resource smoothing of research projects.

Organisation Structure --Principles of organisation and the importance of technology and environment in organisation design.

Group Dynamics--Characteristics and classification of groups, group decision-making and intergroup problems in organisations.

Leadership--Role of leadership in management and leadership styles suitable to Indian conditions.

Conflicts --Types of conflicts that occur in organisation and the methods used to resolve the conflicts.

Organisational Communication--Meaning, concept and importance of organisational structure, communication behaviour, organisation structure's effect on communication, avoiding resentment, principles of effective listening and barriers to listening.

Mass communication --Television in rural communication, steps in developing a TV programme and rural broadcasting.

Course on Human Resources Management (CHRM)

The CHRM is designed to expose the Research Managers to the concepts of human behaviour in organisation and topics related to human resources management in agricultural research. This course is organised for senior scientists heading divisions in the ICAR institutes and professors and heads of departments in the SAUs. The duration of the course is two weeks.

The course was organised twice since January 1, 1990--from August 22 to September 2, 1989 and December 12-23, 1989.

Twenty seven participants attended the course which covered the following topics:

Individual and the organisation; Motivation of the scientists; Interpersonal relationships; Transactional Analysis; Group Dynamics; Leadership; Teamwork; Conflicts; Personnel Policies; Structure and design of Organisation; Organisational Communication; Mass Communication; Research Project Management.

Course on Research Management in Home Science

The main objective of the course is to expose the Professors and Associate Professors in Home Science colleges, Agricultural Universities and ICAR institutes to the basic concepts of developing research projects, application of pragmatic and inter-disciplinary management approach to research problems.

The course was conducted from September 19-23, 1989. Fifteen senior faculty members participated in the course.

The topics covered by the course include :

Research Management Process, Project Planning, Project Evaluation, Basic Elements of Scientific method, Planning for research study, Research Organisational Behaviour, Organisation Structure, Group Dynamics, Leadership, Conflicts, Problem -solving Decision-making, Organisational Communication and Mass Communication.

MANAGEMENT DEVELOPMENT PROGRAMMES

Course on Management Development Programme for Senior Executives of ICAR

The course is organised for senior executives of ICAR, including Deputy Director Generals and Assistant Director Generals of ICAR and Directors of ICAR institutes. This course is designed to strengthen the managerial skills of executives manning the ICAR Headquarters and its various institutes. It also aims at exposing the senior research managers to the concepts of top management functions and relate them to the effective management of agricultural research.

The course was organised from January 30 to February 10, 1989. Thirty senior executives participated in the programme.

The following are the topics covered in the course:

Institution building and models of R&D institutions : National and International experiences.

Strategy Development and Policy Formulation: The Concept and Process of development of research policies and strategies for their implementation.

Project Management : SWOT analysis; Project Selection; Monitoring and Evaluation Methods; Financial Management; Zero-base Budgeting; Budget Formulation and Resource Allocation in ICAR; Management of Financial Resources at the Institute's level.

Human Resources Management: Human Resource Development; Leadership and Group Dynamics; Motivation of Scientists; Interpersonal Relationship; Decision-making Process and Scientist -Administration Interface.

Modernisation of Administration: The concept of MIS; Data Base Management; Computerisation of Office Functions -Scope and Feasibility.

Course on Management Development Programme for Senior Executives of ICAR in Personnel Management

The main objective of the course is to strengthen the management skills of senior executives in the area of 'Personnel Management' by exposing them to the concepts of Human Resources Management with particular reference to management of agricultural research.

This course was conducted from February 13 to 17, 1990. Twenty five senior executives, including Deputy Director Generals, Assistant Director Generals of ICAR and Directors of ICAR institutes, participated in the programme.

SPECIAL TRAINING PROGRAMME

International Training on Library Documentation and Information Management for AFMET, Somalia.

A three months needbased International Training Programme for the Library Professionals of Agricultural Extension and Farm Management Training Project (AFMET), Somalia, was organised from November 1, 1989 to January 30, 1990 at the request of World Bank. The participants were exposed to the management concepts of the scientific libraries and modern techniques in Index to Scientific Reviews (ISR). The trainees were also exposed to the concepts of Selective Dissemination of Information (SDI), Current Awareness Services (CAS), Reprographics/Micrographic services, use of computers for ISR, Users need assessment planning, user education programmes and information consolidation.

TRAINING PROGRAMMES

SL. No.	TITLE OF THE PROGRAMME	DATES	No. OF PARTICIPANTS
1.	Course on Management Development Programme for Senior Executives of ICAR	Jan30-Feb10 1989	30 ✓
2.	Workshop on Effective Communication and Instructional Aids	Jan21-Feb3 1989	33
3.	Programme on approaches to Constraint Analysis and On-Farm Research	Mar13-17 1989	25 ✓
4.	Workshop on Curriculum Design & Development	Apr10-11 1989	18
5.	XXX Foundation Course on Agril. Research Project Management	June6-Oct30 1989	53
6.	Course on Agril.Res.Proj.Management	June13-24 1989	10 ✓
7.	Workshop on Effective Teaching-Learning strategies for UG Vety. Courses	July25-29 1989	28
8.	Workshop on Instructional Tech. and Teaching Effectiveness	Aug1-4 1989	29
9.	XXXI Foundation Course on Agril. Research Project Management	Aug7-Dec29 1989	51

10. Workshop on Methodologies and Approaches to Constraint Analysis in Transfer of Agril. Tech.	Aug8-10 1989	25
11. CASAS-NAARM Asia Regional Seminar on 'Restructuring National Research Systems to meet Agril. Development Challenges'.	Aug22-25 1989	60
✓ 12. Course on Human Resources Management	Aug22-Sept2 1989	19
13. Workshop on Design and analysis of Research Methods	Aug28-31 1989	25
14. Course on Research Management in Home Science	Sep19-30 1989	15
15. XXXII Foundation Course on Agril. Research Project Management	Oct 10-1989 Mar1 -1990	47
✓ 16. International Training Programme on Library Documentation	Nov1-1989 Jan30-1990	2
✓ 17. Course on Agril.Res.Proj.Manage-ment	Nov21-Dec2 1989	18
18. Workshop on Information Tech. and Agril. Information Network	Dec4-8 1989	36
19. XXXIII Foundation Course on Agril. Research Project Management	Dec5-1989 April 30, 1990	27
✓ 20. Course on Human Resources Management	Dec12-23 1989	8
21. XXXIV Foundation Course on Agril. Research Project Management	Feb2-Jun30 1990	20
✓ 22. Course on Management Development Programmes for Senior Executives of ICAR in Personnel Management	Feb.13-17 1990	25

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The various units of the Academy are organising workshops related to specialised topics in their area of specialisation. They are conducted on the basis of research studies undertaken by them. The case materials are developed in advance for the benefit of the participants of workshops.

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The following workshops were conducted since January 1, 1989.

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MEANINGFUL LEARNING AS A COMMUNICATION PROCESS: WORKSHOP ON EFFECTIVE COMMUNICATION AND INSTRUCTIONAL AIDS

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A Workshop on 'Effective Communication and Instructional Aids' was conducted from January 21 - February 3, 1989, to stimulate thinking about teaching and learning from a communication perspective. It was designed to address the issues of creating appropriate environment and organising content resources for greater interaction in the class-room, ultimately making it possible to achieve excellence in teaching-learning.

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Dr. Shirley A. White, Professor, Cornell University, Ithaca, N.Y., U.S.A. was the Facilitator of the Workshop.

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Thirty three faculty members from various ICAR Institutes and State Agricultural Universities, participated in the workshop.

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Recommendations

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The workshop itself was a model of process or experiential learning where the teacher acts as a facilitator -- a counter-model for didactic teaching methods. In the process of experiential learning and close interaction between faculty, facilitator and the participants during the four days of workshop, the following recommendations were made:

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1) Effective communication processes which create meaningful learning experiences should serve as a tool in bringing about change in human understanding and behaviour.

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2) New interactive teaching methodologies and communication technologies must be adopted in the class room teaching for better understanding.

3) The teacher should be oriented and sensitised to their own behaviour in group situation which can give them opportunity to understand what changes they need in order to become more effective communicator in the class room.

4) Individual experiences of both the student and teacher should be actively incorporated in teaching-learning process to achieve higher accomplishment of academic objectives.

WORKSHOP ON CURRICULUM DESIGN AND DEVELOPMENT

A workshop on 'Curriculum Design and Development' was conducted on April 10-11, 1989, for the benefit of the Directors of Summer Institutes. One of the chief purposes of the Summer Institute is to improve teaching capabilities and update courses and curriculum. While emphasising this, the Summer Institute Directors were also exposed to principles of Educational Technology and more innovative and effective teaching techniques. The workshop also discussed various aspects of conduct of Summer Institutes. Eighteen Directors of Summer Institutes participated in the workshop.

Recommendations

A major recommendation that emerged from the deliberations during the workshop was that in all the ICAR-sponsored Summer Institutes, about 15 to 20 per cent of inputs may relate to curriculum development on new arising interdisciplinary frontier subjects and methods of learning-teaching interactions which promote competencies for innovative research and development in the subjects.

WORKSHOP ON EFFECTIVE TEACHING-LEARNING STRATEGIES FOR U.G.VETERINARY COURSES

A workshop on 'Effective Teaching-learning Strategies for Undergraduate Veterinary Courses' was conducted from July 25-29, 1989, in order to expose the participants to the basic process of curriculum design and development at micro level; to help in development of Instructional Objectives, lesson plans and Instructional material to help modify courses which are presently being taught in the Agricultural Universities and to assist in identification of suitable strategies and instructional tools for each topic of the course.

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The workshop was practical oriented, the approach participatory and interactive and the outcome experiential learning. The beneficiaries of the workshop included 28 members of veterinary faculty from different State Agricultural Universities ranging from Assistant Professor to Dean level.

Recommendations

Following are the recommendations of workshop:

- 1) Separate workshops on curriculum development for Undergraduate Veterinary education and on enhancing the Learning-Teaching effectiveness in veterinary colleges need to be organised every year while the workshop on curriculum development may be organised for Deans and Principals of veterinary colleges needs to be organised to regularly for the Associate Professors of veterinary colleges.
- 2) One veterinary college in each region may be identified for organising regular workshops for the teaching faculty in veterinary colleges of the region, taking the technical support from NAARM.
- 3) The veterinary colleges should attempt to introduce structured Learning-Teaching strategies like instructional plan, Experiential processing model and competency-based examination system as these are found to be of great value.

WORKSHOP ON INSTRUCTIONAL TECHNOLOGY AND TEACHING EFFECTIVENESS

A workshop on 'Instructional Technology and Teaching Effectiveness' was conducted from August 1-4, 1989, to familiarise the participants with the broad spectrum of instructional methodology, techniques and approaches so that they can enhance the effectiveness of instructions. During the workshop discussions were also held to develop a suitable methodology for evaluating teaching effectiveness keeping in view of different dimensions of class-room teaching.

Thirty participants from different Agricultural Universities participated in the workshop.

Recommendations

Following are the recommendations of the workshop:

- 1) Every Agricultural University has to have an educational technology cell which could run regular in-service programmes for the teachers of the University on Instructional Designs and Methods.
- 2) In order to develop required manpower trained in educational technology for manning the cells, the Academy develop and offer an Outreach programme for higher education in Agriculture.

WORKSHOP ON METHODOLOGIES AND APPROACHES TO CONSTRAINT ANALYSIS IN TRANSFER OF AGRICULTURAL TECHNOLOGIES

A workshop on 'Methodologies and Approaches to Constraint Analysis in Transfer of Agricultural Technologies' was organised from August 8-10, 1989. The experience of the Training Programme on Approaches to constraint analysis and On-Farm-Research organised during March 1989 constituted the basis for organising the workshop. The workshop was organised with a view to examining the methodologies and approaches to constraint analysis, learning from case studies the mechanics of constraint analysis, discussing ways and means of amelioration of constraints and identifying possible causes for constraint analysis to TOT projects.

Twenty five participants drawn from various ICAR Institutes and State Agricultural Universities participated in the workshop. The workshop was able to impart training on methodologies, methods of conducting case studies and overcoming the constraints through conscious efforts. It also came out with recommendations to first line extension projects carried out by National Agricultural Research Systems.

WORKSHOP ON DESIGN AND ANALYSIS OF RESEARCH METHODS

A workshop on 'Design and Analysis of Research Methods' was conducted from August 28-31, 1989, with the help of Dr. Larry E. Miller as Facilitator. The workshop aimed at broadening the perspectives of Senior level Scientists and opening a window to an awareness of the contributions various research methods can make in the area of social sciences research. Some of the major points that emerge from the deliberations in the workshop are :

1) In order to establish credibility and make significant contribution to the overall agricultural research efforts, social scientists need to acquire the required knowledge and develop sufficient skills to undertake the extremely complex and difficult social science research.

2) The usefulness of well-executed research methods of various kinds, even if they are simple, should be properly understood and appreciated by the researchers.

3) Researchers are required to develop specialised skills for developing effective questionnaires while conducting survey research.

4) Highly specialised skills are required to develop new instruments or to adapt the existing ones with sufficient validity and reliability for measuring the attitudes.

5) Conducting interviews is an art by itself which the researcher should learn through training and by experience, but should never be considered as something that anyone can do.

6) In addition to normal correlation and regression techniques, researchers should also be familiar with multivariate statistical tools like canonical correlation, discriminant function analysis, multicollinearity, multivariate analysis of variance, etc.

7) While dealing with qualitative data in social sciences, researchers should develop thorough insights into various analytical techniques including interpretation to make valid conclusions, and be aware of their limitations.

8) Utmost care should be exercised by the researchers while undertaking experimental research in social sciences, to remove the external as well as the internal threats to validity through employing suitable designs and appropriate analytical techniques.

9) The usefulness and application of non-parametric statistical techniques in social science research as complementary to parametric techniques should be well understood and appreciated by the researchers.

Twenty five senior level scientists from the Agricultural Scientists and ICAR Institutes participated in the workshop.

The workshop was evaluated through a pre and post workshop measurement of the participant perception of their level of competence in the respective area of

content of the workshop. The participants were satisfied with the overall content, coverage, organisation and accomplishment of the workshop.

WORKSHOP ON INFORMATION TECHNOLOGY AND AGRICULTURAL LIBRARIES INFORMATION NETWORKS

A workshop on 'Information Technology and Agricultural Libraries Information Network' was organised from December 4-8, 1989. The workshop discussed the issues relating to the adoption of the technologies in modernising the libraries and serving the clientele more effectively.

Thirty six participants representing various ICAR Institutes and the State Agricultural University Libraries participated in the workshop. Mr. William B. Feidt, Leader, Information Technology Group, National Agricultural Library, Beltsville, Maryland, U.S.A., participated as observer.

A major outcome of the deliberations in the workshop was an immediate need for establishing a National Network of Agricultural Libraries in order to share the resources and exchange the information.

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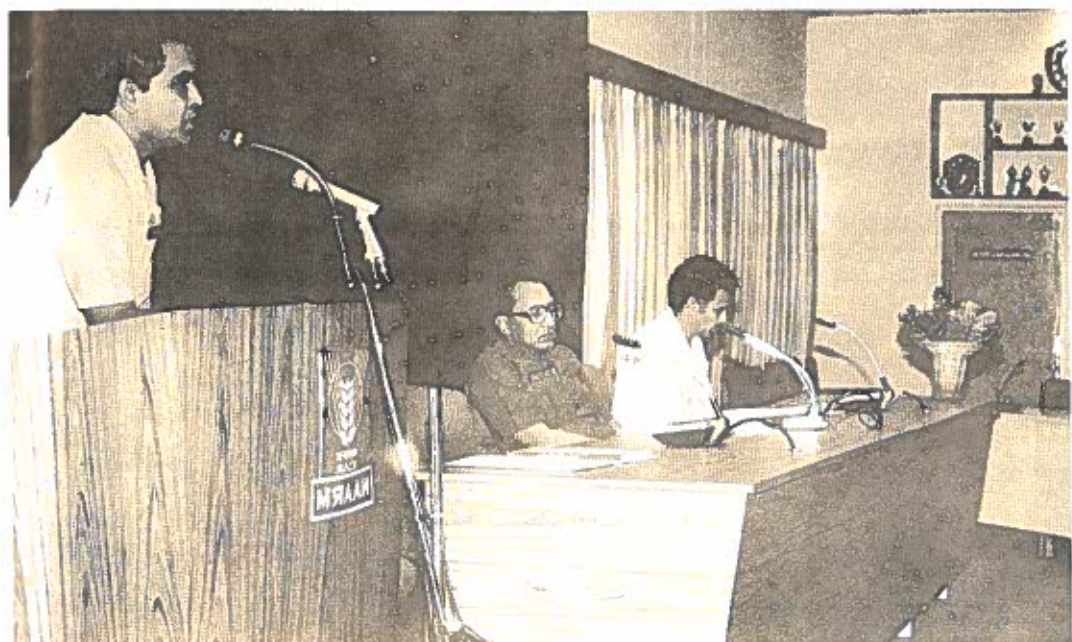
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Dr.K.V.Raman, Director, addressing the participants of the workshop on "Information Technology and Agricultural Information Network".



Dr.K.P.C.Rao, Coordinator, addressing the participants of the workshop on "Methodologies and Approaches to Constraint Analysis in the Transfer of Agricultural Technology"



Dr.K.V.Raman, Director, NAARM and Chairman, Seminar Organising Committee welcoming the delegates



A section of the delegates

2.4 CASAS-NAARM ASIA REGIONAL SEMINAR

The Commonwealth Association of Scientific Agricultural Societies (CASAS) in collaboration with the National Academy of Agricultural Research Management (NAARM), Rajendranagar, Hyderabad, organised a four-day Asia Regional Seminar at the Academy from August 22-25, 1989 on 'Restructuring National Research Systems to meet Agricultural Development Challenges'.

The seminar focussed attention on the attempts being made by the participating countries to restructure the agricultural research system keeping in view the national priorities and their future requirements. Sixty scientists, including delegates from Bangladesh, Brunei, Malaysia, Singapore and India participated in the seminar. Her Excellency Smt. Kumudben Joshi, Governor of Andhra Pradesh was the chief guest. Inaugurating the seminar, she pleaded for accordng the status of industry to agriculture to give impetus to agricultural development.

Delivering the keynote address, Dr.M.S.Swaminathan, eminent agricultural scientist and Honorary Director of the Centre for Research on Sustainable Agriculture and Rural Development, Madras, said the nature of food security challenge has shifted from physical access to one of economic access to food. Diversification of employment opportunities in rural areas through integrated attention to the primary, secondary and tertiary sectors of employment and income generation has therefore become urgent necessity.

Dr. Abdul Halim Hassan, Dy. Director-General, Palm Oil Research Institute, Malaysia, and Dr. N.S. Randhawa, Director-General, ICAR, participated in the programme.

Recommendations of the Seminar

A POLICY

The evolution of any policy mix would be based on past experience of development and performance evaluation of the same at a given stage. Such an analysis would provide a bench mark and a backstop against which future policy could be formulated. The policy would naturally take into account several challenges facing agriculture like the shifting of food security challenge from physical access to economical and ecological access to food and the need to provide more employment opportunities in the farm sector. The policy should

centre around research prioritisation, decentralised research planning and provision of adequate resources for ongoing research as well as anticipatory research.

2. In view of several alternative production possibilities, research prioritisation should take into account the national mandate and regional variations in resource endowments to ensure optimum use of total available resources.
3. Decentralised research planning will help evolve planning process in floor-to-top fashion, in keeping with felt needs and domestic resource capabilities, operating from micro to macro level.
4. There should be sufficient technical and financial resources to prevent biological impoverishment. Adequate financial resources to fulfil the mandate should be provided to research systems at all levels and the phenomenon of surplus trained manpower on the one hand and several unfilled positions on the other should be guarded against by cutting out delays in recruitment and postings.
5. All developing countries depend rather excessively on public sector research efforts. Involvement of the private sector in at least economically viable entrepreneurship areas should be encouraged. It would be wise for the Government to concentrate on basic and fundamental research requiring massive financial outlays, and to allocate jobs/areas like seed multiplication, implement production and technology generating components to private research organizations. However, such efforts should be effectively integrated and monitored as a subsystem of the national research system.

While individual excellence in research is to be encouraged, institutionalising such excellence will yield maximum results. This could be facilitated by forging professional links and encouraging team work among scientists in various institutions involved in Agricultural research. Institution-building should no longer be viewed as brick and mortar and increasing financial outlays. An institution of research should breathe the very spirit of academic excellence and there should be an atmosphere of team work all round. Since research is a creative activity, institutions should not be allowed to become obsolete, but should be fostered with vision and foresight to maintain sustainable dynamism.

B. RESEARCH

1. Agricultural research should strive to improve the livelihood security of the poor, and enrich the quality of life of the farmer, while achieving the quantitative targets set in its nation's policies and programmes.

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2. With the increase of population pressure on land, unsustainable use of natural resources, destruction of ecosystem rich in biological diversity, and increasing soil and gene erosion, research should aim to integrate the best in frontier technologies with the best in traditional practices. It should shift from an imitative to a creative R & D system which should promote commitment in its staff and their continuing retraining.

3. Field problems should be solved by multidisciplinary teamwork based on a systems approach. Agriculture research should manifest a two-dimensional growth with scientific research being carried on in labs and technological research in the farmers' fields, both of them integrally related.

4. The emerging research systems should adopt a progressive personnel policy by attracting new blood scientists to work in frontier areas of science and also developing outstanding scientists to work in ecologically handicapped areas.

5. Technical information exchange among scientists should be encouraged by ensuring their participation in National/Regional level workshops. Separate budgets should be available for this purpose.

6. The apex bodies dealing with research in General Universities may develop ways and means of promoting systematically collaborative upstream research in areas like Bio Technology, Ergonomics, processing of foods, nutrition, space imaging, computer applications and informatics. There should be a futurology group monitoring trends in changes of ocean levels and other climatic changes, for the benefit of future generations.

7. The research system should be sufficiently equipped to bridge the gap between 'know-how' and 'do-how' between 'Lab and Land'.

(C) MANPOWER PLANNING (HUMAN RESOURCES DEVELOPMENT) THROUGH EDUCATION

1. It is observed that the research output in some of the developing countries does not measure up to the cherished goals, because of the low trained manpower employed in their national research systems. It is therefore, recommended that efforts in NARS should be geared up by a three pronged strategy viz. domestic skill upgradation activities, skill upgradation through international collaboration and by creating working conditions fostering trust and confidence among the various levels of scientists.

2. A research system should plan its manpower requirements realistically on the basis of available and not utopian manpower availability.

3. The content of agricultural education has to be conditioned by current and future needs of research, in order to maximise research output at any point of time.
4. Agricultural polytechnics may be started, offering a two year post-secondary diploma in Agro-production in the regional language to produce midlevel technicians to support Farming Systems Research.
5. Post Graduate Courses may be provided in interdisciplinary areas of research like Bioenergy, Biotechnology, Environmental Science and Farming System and Management.
6. Similarly, U.G. Curriculum may be restructured, to introduce new courses and to modify existing ones that will focus attention on sustainable agriculture and socioecological well-being.
7. Agricultural Subjects should be introduced in the School Curriculum, and Science (relating to health, foods, biology, ecology, chemistry and physics) may be taught in an integrated manner in terms of location specific plant, animal and tree kinds, their uses and needs for conservation and production.
8. The instructional system in Universities should be built around case study approaches, experiential learning in the form of group projects, field problem based group discussions, poster sessions, exploration of micro environment for recording interdisciplinary perceptions and group learning exercises.
9. The emerging research needs warrant continuous upgradation of competencies, attitudes and orientation of teachers in Agricultural Universities. They may be required to undergo a structured training in an organisation once every five years.
10. In order to offer the required upgradation training programmes a national level institution may be identified/developed and supported to undertake the required research and to offer training on all relevant areas like curriculum development, institutional systems, and development of field-based learning resources.
11. Institutionalized training may be imparted at the orientation stage and also subsequent retraining stages. The training should include organisational behaviour areas like motivation, leadership, conflict resolution besides professional skills and subject competencies,etc.

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12. Incentives may be provided to research staff to work in remote areas on the lines provided for defence personnel. Facilities may be provided to staff for sabbatical leave to upgrade their skills and knowledge in consonance with the objectives of the sponsoring institutions.

D RESEARCH AND EXTENSION

The evolution and existence of Technology by itself is not a sufficient condition for its use as an end product. Education and extension should take over from that stage and serve as effective vehicles to carry the research to the end users. Sometimes, based on the feedback, even the technology needs to be modified by NARS while drawing up first line extension projects. Social Scientists and Agriculture extension workers should play a big role in spreading the knowledge and practices of improved agriculture to farmers. Thus NARS should establish forward and backward linkages with education systems. These linkages serve to provide mutual complementarity and strength for the research and extension systems.

1. The Agricultural Research and Extension linkages must be institutionalised to meet the present challenges. This could be done by associating the extension machinery at the initial stage of research formulation and at the stage of releasing the technology. Feedback mechanism between Farmers and Researchers should be stronger so that technology generation becomes appropriate as per needs of the clients.
2. The gap between the technology generation and utilisation should be narrowed down by bringing the functions of both the research and extension systems under one umbrella.
3. The extension system should adopt participatory approach in developing the action programme by involving input supply system. This will ensure the supply of inputs required for adoption before the release of the technological package.
4. The Extension and Research Systems should be oriented towards each other so that they can function effectively and efficiently keeping in view the problems of farmers. A system should be devised to impart training to the personnel of Extension Systems in understanding the sophisticated technology to be imparted to the farmers.
5. Emphasis should be given to research on extension to enhance the effectiveness of the system. Priority research areas should include extension methodologies and development communications in planning, organizing and

conducting such research endeavours. The research on the extension system should also include inputs from social scientists.

6. Research system should establish a strategy where frontline extension workers are able to understand the field problems and obtain feed-back regarding the technological requirements which should be channelled to national extension service programmes.

7. To cope with the complex socio-cultural situations in the transfer of sophisticated technologies, extension system should be equipped with professionally trained and competent personnel. They must be able to provide dynamic leadership so that extension programmes are smoothly and effectively implemented.

2.5 CONSULTANCY SERVICES

The Academy is providing consultancy service in the area of Agricultural Research Management and Educational Technology. The Academy is guiding the State Agricultural Universities and ICAR institutes in conducting training programmes in these areas. The Faculty members of the Academy are specialised in various sub-disciplines and deliver special lectures in institutions which organise training programmes to its own staff. These institutes have found the participation of NAARM faculty immensely useful.

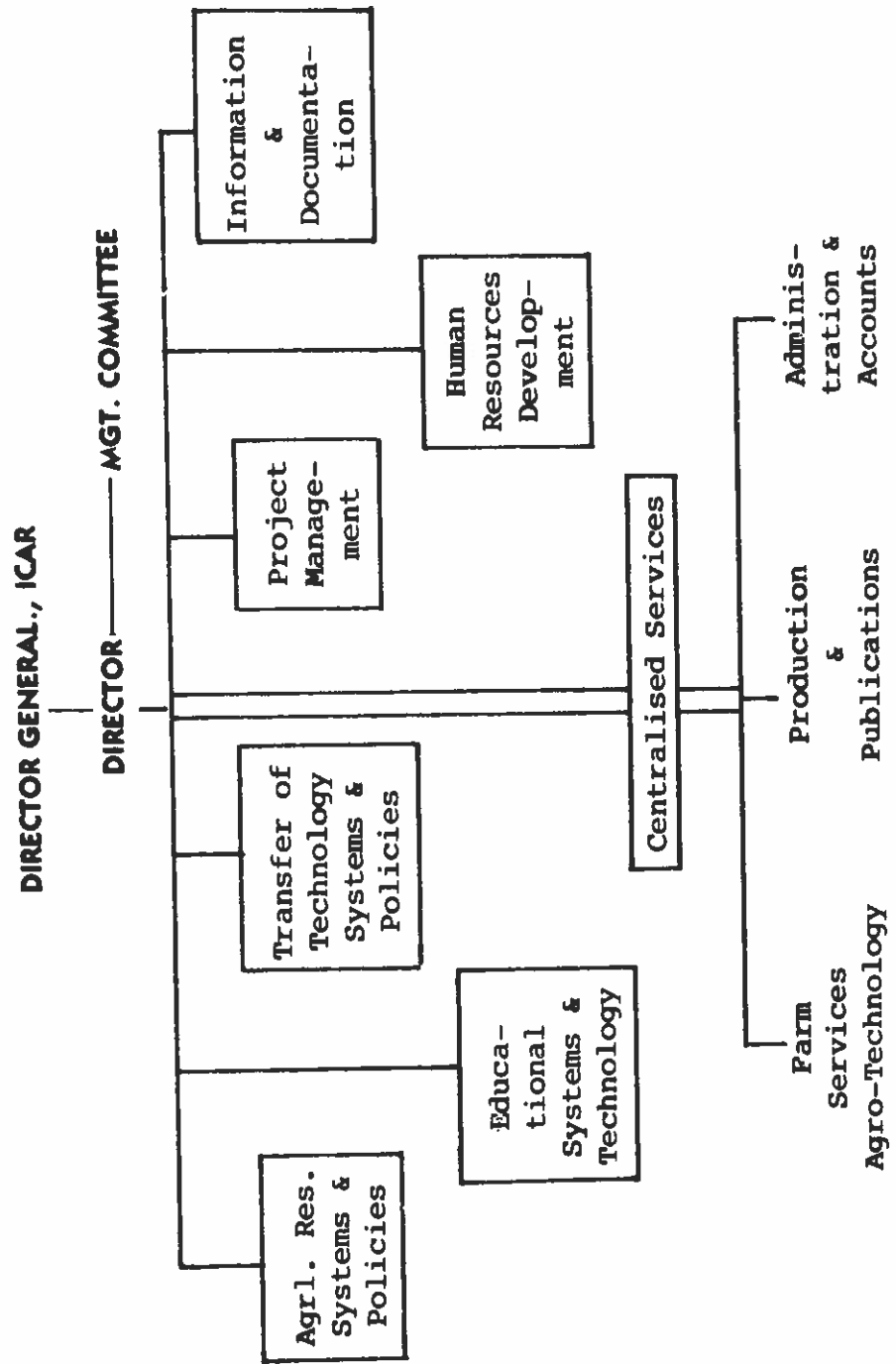
3. ORGANISATIONAL STRUCTURE AND SALIENT ACTIVITIES

The Academy is organised into six subject-matter oriented functional units with a view to imparting effective training and conducting research studies in a meaningful way. These are:

1. Agricultural Research Systems and Policies
2. Educational Systems and Technology
3. Transfer of Technology Systems and Policies
4. Project Management
5. Human Resource Development
6. Information and Documentation

All the above mentioned units function with a strong support from Centralised Services comprising
Farm Services and Agro-Technology Blocks
Production and Publications
Administration and Accounts

ORGANISATIONAL STRUCTURE



3.1 AGRICULTURAL RESEARCH SYSTEMS AND POLICIES UNIT

Development of Agricultural Sector is conditioned by resilience and capacities of agricultural research system. For, the functions relating to technology generation, gathering of information and procedures involved in agricultural research system would be used jointly for agricultural education, training, extension and client support to bring about desired changes in development strategies. The Agricultural Research Systems and Policies Unit studies the different systems of agricultural research in order to strengthen the various research systems that are essential for agricultural development.

Objectives

- * to study issues on prioritisation of research in Agricultural Research Systems and to assist in development of appropriate science policies,
- * to build agricultural resource and research data bases for developing suitable research plans and efficient management of National Agricultural Research Systems,
- * to develop a systems approach integrating indigenous knowledge with relevant technologies for varied farming situations and constraints,
- * to study National as well as International Agricultural Research Systems and suggest suitable policy alternatives for improvement,
- * to undertake studies on the socio-economic impact of resource allocation and investment in agricultural research, and,
- * to undertake studies on futurology in agriculture and technological forecasting and assessment.

ACTIVITIES

Research Projects

1. "Developing suitable Management Information Systems (MIS) for the National Agricultural Research System"
(T. Balaguru and K. Vidyasagar Rao)

This project was initiated to develop an organised MIS to serve the research managers in the ICAR and Agricultural Universities Systems as an important management tool to efficiently manage their research activities, human resources, funds and other physical facilities. Various proformae have been developed to collect data on research projects, personnel, funds and physical facilities.

2. "Evolving a Suitable Methodological Framework for priority setting in Agricultural Research".
(Jagdeesh C. Kalla and T. Balaguru)

As research expenditures rise and resources are not forthcoming, unconstrained, research managers are hard-pressed for more objective and systematic methods for prioritising research in agriculture. This project aims to develop an appropriate methodological frame-work for research prioritisation. Action has been initiated to develop relevant proformae to collect data from the policy makers, research managers and scientists.

3. "Patterns of Research Resource Allocation in Agricultural Universities".
(K.P.C.Rao and Jagdeesh C. Kalla)

This project has been undertaken to find out the present pattern of allocation of research resources among disciplines\commodity groups in different Agricultural Universities and their Regional Research Stations in the country and to suggest changes in resource allocation among disciplines\commodity group based on some sound allocation criteria.

4. "Assessment of returns to expenditure on research and development of production technologies for rice-fallow pulses and chillies in Andhra Pradesh".
(K.P.C. Rao)

This project has been undertaken to estimate the rate of return on investments made on specific commodity research programmes and measure the impact of research and extension on the productivity of pulses in Andhra Pradesh.

5. "Study on Scientific Manpower Planning and Utilisation in ICAR Research System".
(VE. Sabarathnam)

The following subprojects of the project were completed during the year under report :

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- a. Estimation of Scientific manpower efficiency
- b. Time Management analysis for Scientific Manpower
- a. "Estimation of Scientific manpower efficiency

The following were the objectives of this sub-project.

- 1) To estimate the efficiency of scientific manpower in different disciplines of ICAR system.
- 2) To analyse the factors responsible for efficiency variation in scientific manpower.

Data from 105 scientists in four major institutes of ICAR (two crop science institutes and two animal science institutes) were collected and analysed statistically.

Salient findings are:

- 1) Majority of the Scientists in the sample institutes came under the average category of efficiency in scientific research.
- 2) Considerable percentage of scientific manpower was poor in efficiency.
- 3) Highly significant variation among different institutes with regard to scientific manpower efficiency was observed in the study.
- 4) The most important indication of scientific manpower efficiency was found to be feasibility of research findings under user's situation.
- 5) The variables which had significant positive relationship with scientific manpower efficiency were a) nature of involvement of the scientist in the research project, b) distribution of research time for projects, c) self perception of the scientist about scientific capability and d) maintenance of personal file.
- 6) The number of research projects which a scientist had at a time showed negative correlation with the scientific manpower efficiency.
- 7) The most important variables that contributed to scientific efficiency, as per the study, were a) nature of involvement of the scientist as a leader or associate in the project b) perception about self-scientific capability, and c) the number of research projects handled at a time.

- 8) The analysis variance indicated that there is significant difference between the disciplines with regard to manpower efficiency.
 - 9) The variation between the institutes with regard to scientific manpower was more in animal science institutes than in plant science institutes. But, the effect of disciplinary variation on efficiency was high in plant sciences compared to animal science disciplines. The effect of interaction between the institutes and disciplines is again slightly more in plant science institutes than in animal science institutes.
- b. "Time Management Analysis for Scientific manpower".

The objective of the sub-project was to analyse the ways in which the official time was utilised by scientific manpower of ICAR research time.

Salient findings:

- 1) Significant amount of variation was observed among the scientists with regard to the official time spent on research work. It varied from 20 to 100 per cent.
- 2) Significant intra-institutional difference with regard to the different ways of distribution of research time for different projects by the scientific manpower was observed.
- 3) Inter-institutional difference regarding distribution of research time for different projects by the scientists was insignificant.
- 4) Time spent on training and on teaching activities by scientists differed significantly between one scientist and another.
- 5) Majority of the manpower reported to have spent only less time of their day's time in the research plots.
- 6) Most of the scientific manpower had spent only less than an hour in a day in the library.
- 7) Majority of the scientific manpower could not account the hours spent in a day on an average in the research plots.
- 8) Considerable percentage of scientific manpower had no information regarding the time spent on an average in the laboratory.
- 9) Scientific manpower in ICAR system had spent from three to 12 days in a

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month for writing research papers. On the other hand, majority had no idea regarding the time they spent in treating and disseminating their findings to users and promoters.

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6. "Policy analysis in involvement of Technology- Advocates in development of crop technologies"

(VE.Sabarathnam)

This project aims at finding out the role the technology -advocate can play at each step of technology development process and assessing problems involved in it.

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7. "Estimation and Evaluation of Students' performance at different levels".

(K.Vidyasagar Rao)

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Data on grade point averages and other socio-economic factors of B.Sc (Ag). students for four batches (1981-84) of APAU, Rajendranagar, Hyderabad, have been collected and analysed.

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8. "Development of Database on Crops, Forestry, Animal and Fishery Resources in National Agricultural Research Systems"

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(K.Vidyasagar Rao and P. Kumar)

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The project aims at analysing and documenting long-term trends in the constituents of database on crops, livestock, forestry and fisheries. It is expected to provide basic information for policy formulation at national and regional levels. It also helps to identify and delineate the elements of future strategies for optimisation of agricultural production.

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WORKSHOPS

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1. **Workshop on Methodologies and Approaches to Constraint Analysis in Transfer of Agricultural Technologies**

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A workshop on Methodologies and Approaches to Constraint Analysis in Transfer of Agricultural Technologies was organised from August 8-10, 1989. Dr.K.P.C.Rao coordinated the workshop.

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2. Workshop on Design and Analysis of Research Methods

A workshop on Design and Analysis of Research Methods was organised from August 28 to 31, 1989, for senior scientists of ICAR institutes and State Agricultural Universities. Dr.T.Balaguru coordinated the workshop and Dr. Larry E. Miller, Professor of Agricultural Education, The Ohio State University, U.S.A., acted as the Facilitator.

Course Material Developed

1. Economic evaluation of agricultural research projects
2. Agricultural and economic development in five-year Plans
3. Constraint analysis - A case study of grassland management system in arid areas of Western Rajasthan
4. Intersystem approach for Technology Development
5. Management Information System
6. Farming System Development
7. Research Prioritisation in Agricultural
8. Livestock production potential & constraints
9. Recent trends in research : Livestock management
10. Constraint analysis in research and adoption
11. Land systems and reform measures
12. Agricultural labour & employment
13. Basic concepts of linear programming

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**EDUCATIONAL SYSTEMS AND
TECHNOLOGY UNIT**

The primary function of this unit is to render assistance to Agricultural Universities and ICAR institutes dealing with education and provide support for development of better teaching techniques. Education technologists interact closely with Subject Matter Specialists to develop viable and useful courses and teaching techniques.

Objectives

- * to organise teaching and training programmes in higher agricultural education with special reference to curriculum development, instructional systems, production of learning resource materials, evaluation and assessment, and educational administration,
- * to undertake research projects and case studies for evolving more effective and efficient approaches, methods and models in higher agricultural education system,
- * to develop information data base on education systems in SAUs and ICAR institutes,
- * to provide technical support to SAUs and ICAR institutes on educational matters to enhance their teaching-learning effectiveness, and,
- * to establish linkages with international bodies imparting higher agricultural education,

ACTIVITIES

Workshops

1 Workshop on Curriculum Design and Development

A workshop on Curriculum Design & Development was organised from April 10-11, 1989 for the Directors of Summer Institutes from ICAR institutes and Agricultural Universities. The workshop was coordinated by Dr. Devesh Khosla.

2. Workshop on Effective Teaching-Learning Strategies for Undergraduate Veterinary Courses

A workshop on Effective Teaching-Learning Strategies for Undergraduate Veterinary Courses for Professors and Associate Professors of Agricultural Universities offering undergraduate Veterinary courses in Basic, Para Clinical and Production areas from July 25-29, 1990. Dr. K.M.Reddy coordinated the workshop.

3. Workshop on Instructional Technology and Teaching Effectiveness

A Workshop on "Instructional Technology & Teaching Effectiveness" for Deans, Associate Deans, Directors of Instruction, Principals, Senior Professors of the Agricultural Universities and ICAR Institutes was conducted from August 1-4, 1989. Dr. Devesh Kishore coordinated the workshop.

Courses

Course on Research Management in Home Science

A course on Research Management in Home Science was organised from September 19-23, 1989 for Professors and Associate Professors in Home Science Colleges and Agricultural Universities and ICAR institutes. Dr. Devesh Kishore coordinated the course.

Research Projects

1. "Production of learning centred resource materials - Video lessons"

(Dr. Devesh Kishore, Dr. R. Sudarsanam, Dr. K.M.Reddy and Dr. Gopalam)

The project aims at producing video lessons and structured learning materials as effective learning tools. Initially, packages will be developed in areas of Communication, Information Processing, Computer and Biochemistry.

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2. "To develop and offer out -reach programme in educational technology for agriculture"

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(Dr. R. Sudarsanam, Dr. Devesh Kishore, Dr. K.M.Reddy, Dr. A.Gopalam,
Dr. C. Jayannadham and Shri R.B.Gaddagimath)

The project aims at developing and offering a distance education programme in educational programme. The objectives of the project include: developing an operational scheme and a curriculum for two semester diploma programme in educational technology and developing suitable teaching-learning methods including those meant to be used in 'contact programmes' for the participants of the programme.

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3.3 TRANSFER OF TECHNOLOGY SYSTEMS AND POLICIES UNIT

Development of appropriate transfer of technology systems is important for improving the productivity of both the researchers and farmers. The effectiveness of Transfer of Technology systems depends on various factors such as managerial efficiency of the managers, span of control, communication strategies, planning and execution. Further, the informal and institutional linkages are the key determinants of success of the TOT systems. Identifying these linkages is a major mandate of this Unit.

Objectives

- * to study the problems and programmes of technology generation and evolve tools for this assessment in transfer of technology,
- * to identify the managerial constraints in existing systems and develop suitable and integrated approaches for effective transfer of technology,
- * to suggest suitable policy framework for effective implementation of technology, and,
- * to evolve an efficient agricultural communication system to support the aforesaid activities,

Research Projects

1. "Study on Impact of Agricultural Research Project Management Training on ICAR Scientists and Their Training Needs"

(K.V.Raman, R.K.Samanta, R.Chittiraichelvan and A.U.Gadewar)

Objectives:

- * to evaluate the impact of NAARM training on ICAR Scientists and their environment,
- * to find out how much has the NAARM training helped the Scientists perform their jobs better in their working situations,

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* to find out how much the client institutes have benefited from their Scientists trained at NAARM,

* to document the strengths and weaknesses of existing training programmes, and

* to evaluate the training needs of the ICAR Scientists

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Salient findings of the study are:

1) The trained scientists have better appreciation and understanding of the Agricultural Research Project Management (ARPM) training than their counterparts who have not been trained in any of the training programmes. Therefore, all categories of scientists irrespective of cadre in ICAR must undergo ARPM training at the Academy which may help them in understanding and managing agricultural research projects better.

2) ARPM training has been able to induce team spirit, leadership and fraternity among the scientists. This significant aspect can be exploited suitably by the Institutes for increasing research productivity of both the scientists and institutes to bring overall institutional development.

3) The Field Experience Training Programme should receive higher importance and emphasis in its design, content and implementation.

4) The existing evaluation system in the Foundation Course may be modified suitably to include continuous and effective monitoring of the trainees over the entire period of the training programme.

5) Post-training evaluation and follow-up of the Scientists should be done at the concerned institute level and at the Academy to get the feedback to improve the training programme and for better utilisation of Scientists' skills for research productivity and institutional development.

6) Course curriculum and the contents of various topics require modifications to keep up the pace and relevance with the changing needs of the scientists and institutes.

7) Due care may be taken in recruitment and development of NAARM Faculty considering the needs and suitability of the positions and nature of the job.

8) A conducive climate and infrastructural facilities may be created in all the ICAR institutes so that things learnt in ARPM training at the Academy can be

utilised and applied by the scientists successfully, meaningfully and more effectively.

9) In dealing with management topics like research project formulation and implementation, the network analysis and management process, examples may be drawn from Indian experience with agricultural bias with the support of case studies.

10) ARPM training should be need-based and relevant to the working situation of the scientists working in various projects and locations of ICAR institutes.

2. " Research Management --Utilisation interface: Case study on Oilseeds"
(K.V.Raman and R.K.Samanta)

Objectives :

- * to study the adequacy and variety of criteria applied for formulation of research programme to the farmers and requirement priorities, market structures and policies

- * to give insight into the wide knowledge-practice (technology adoption) gap and develop strategies to increase the research output to farmers' requirements

- * to study the managerial constraints in effective research, extension system and linkages that will promote smooth transfer of technology information and its adoption

- * to evolve effective instructional structures for R&D and extension linkages

3. "Documentation of Various Indigenous Knowledge Systems in
Agriculture"
(R.Chittiraichelvan, R.K.Samanta, A.U.Gadewar and Jagdeesh C.Kalla)

Objectives:

- * to review and document various indigenous knowledge systems in the area of agriculture and allied fields

- * to find out cost-benefit ratios of indigenous technical knowledge systems

- * to assess the feasibility of incorporating indigenous

knowledge in technology generation ,transmission and utilisation.

4. "Evaluation of Use and Impact of Sorghum Publication--Nutritional Disorders of Sorghum"
(K.V.Raman and A.U.Gadewar)

Objectives:

- * to find out the extent of distribution of Sorghum publication through out India with particular focus on Sorghum growing areas
- * to find out who are recipients of publication
- * to ascertain its usefulness/impact among the researchers, professionals in teaching and extension personnel.

Work on the last three projects mentioned above is in progress.

Workshops

A workshop on Effective Communication and Instructional Aids was organised from January 31 to February 3, 1989. Dr. R.K. Samanta coordinated the Workshop. Dr. Shirley A. White, Professor, Cornell University, U.S.A. was the Facilitator.

Course Materials/Lecture Notes developed

- 1.Development Communication in Agriculture
- 2.Technology Transfer and Appropriate Technology
- 3.Rural Development: Philosophy and Basic Issues
- 4.Effective Presentation Techniques
- 5.Rural Leadership
- 6.Role Play Exercises
- 7.Audio Visual Aids for Effective Communication
- 8.Instructional Structure for Technology Transfer
- 9.Pollution
- 10.Systems Approach to Technology Development
- 11.Diffusion of Innovations
- 12.Consequences of Technology
- 13.Treatment of Research Findings: Theory and Practical
- 14.Mass Media Communication

3.4

PROJECT MANAGEMENT UNIT

Project Management Unit is designed to integrate complex efforts with regard to planning, scheduling, directing and controlling Institute's resources for a project established with specific goals and objectives. This Unit provides an individual research worker the necessary tools and techniques which can help him find proven and workable solutions to problems that may occur in the management of a project.

Objectives

- * to evolve an appropriate methodology for identifying relevant Agricultural Research Projects in a given thrust area and to develop criteria for their appraisal,
- * to develop and formulate a holistic/systems approach for project planning and management,
- * to study issues, develop case studies and to evolve suitable tools for carrying out the activities related to project cycle more effectively, and,
- * to develop a data-base system on Agricultural Research Projects needed to fulfil the above objectives.

ACTIVITIES

The Unit has participated in conduct of training programmes for middle and senior-level scientists and foundation courses for direct recruits in Agricultural Research Service in the area of Project Management. It has undertaken case studies to improve the management of All India Coordinated Research Projects and National Agricultural Research Projects in order to improve the mechanism of implementation, monitoring and control of the projects.

Research work is being undertaken in the areas of Project Selection, Management of Research Stations through the use of Project Management Techniques/Networks. Studies are also being undertaken in procedures for the implementation of project-wise budgeting in ICAR institutes.

Course

Course on Agricultural Research Project Management

This course was organised twice, from June 13-24, 1989 and from November 21-December 2, 1989, for senior faculty members and senior scientists of ICAR institutes. Dr. V. Raghavendra Rao and Dr. K. Venkateswarlu were the Course Coordinators.

Research Projects

1. "Development of Appropriate Criteria for Project Appraisal in Basic & Applied Research in Agriculture"

(V. Raghavendra Rao)

Work on this project has been initiated.

2. "Project Management Techniques in Research & Development for Agriculture, Veterinary and Fisheries"

(K. Venkateswarlu)

A major part of the work on this book is completed. Some mathematical concepts like the Linear Programming are being incorporated.

3. "Project-wise Budgeting in ICAR Institutes"

(P. Manikandan)

A comprehensive booklet titled 'An Approach to Project Budgeting Technique' was prepared detailing the features of the proposed Project Budgeting System. The booklet was distributed to various ICAR Institutes, Project Directorates and National Research Centres to enable them to adopt this system with suitable modifications that may be necessary to suit the requirements of individual institutes. The essential features of the Project Budgeting System are:

1) Since one of the objectives of the proposed system is cost control, for budgeting purposes, costs should be classified and budget levels established on the basis of twin factors of controllability and traceability. The principle behind the classification of costs is that the costs should be budgeted only by those

levels that are in the best position to control them. On the basis of the factors of controllability and traceability, costs are classified as follows:

- a) Costs traceable to projects and controllable at project level.
eg. Chemicals, travel etc. directly related to project
- b) Costs traceable to project but not controllable at project level
eg. Land preparation and planting costs
- c) Costs neither traceable to project nor controllable at project level.

This cost is further classified into three levels:

- i) traceable to and controllable by divisions
eg. Casual labour, common material cost
- ii) traceable to but not controllable by divisions
eg. Scientists' salaries
- iii) neither traceable to nor controllable by divisions
eg. a great proportion of the purely administrative expenses.

2) The responsibility for preparing budgets is to be determined on the basis of cost controllability. Accordingly, three budget levels are proposed, viz., Project, Division / Service Centre and Institute. Each level will budget only for those costs that are controllable at that level. The budget and the level at which it should be prepared are indicated below :

<u>Budget</u>	<u>Budget Level</u>
a. Service Centre Budget	Service Centre
b. Project Budget	Project
c. Division Budget	Division
d. Other costs Budget	Institute
e. Equipment Purchase Budget	Institute

3) Currently, equipment cost is budgeted for and booked to specific projects. However, equipments are used for more than one year and for many projects. Therefore, it is proposed that

- a) Equipments should be budgeted for at the Institute level
- b) Equipment cost to be amortised over its estimated useful life, and
- c) Annual amount of amortisation to be allocated to Projects by Division

head / Director depending on whether the equipment is used exclusively by one Division or by more than one Division.

Amortisation and allocation to projects are proposed mainly for project costing purposes.

4. In order to make the system useful, control reports are recommended for use. These control reports measure performance against targets. The following table gives the details of control reports proposed and their frequency.

Report	Frequency	Distribution
1. Service Centre Expenditure Report	Monthly	Service Centre Head, Director
2. Project Expenditure Report	Monthly	Project Leader
3. Summarised Project Expenditure Report	Quarterly	Director, Project Leaders concerned
4. Exception Report on Project Expenditure	Half-Yearly	Director Project Expenditure
5. Division Expenditure Report	Monthly	Division Head Report
6. Summarised Division Expenditure Report	Quarterly	Director Expenditure Report
7. Other Costs Report	Monthly	Director
8. Activity Report	Quarterly	Director

Exception report on project expenditure facilitates management by exception. This report will only list out projects which have utilized 10% and less than 10% of the budget and those which have utilized more than 90% of the budget by the end of six months. This report will enable the Director to re-allocate financial resources among projects.

5) It is also suggested that as far as possible, the technical programme for the subsequent year be finalised by September itself in order to facilitate a realistic

projection of budget estimates. The system also recommends considerable delegation of powers for spending to the Heads of Divisions. The system proposes the use of 13 different forms that will help in the easy adoption of the proposed project budgeting technique.

Computerized system for
project budgeting
accounting for resources
management at the project level

3. Scarcity of equipment should be
considered as a project activity
in the project budget
and 20% of the total

to be budgeted at the project level
for purchase of other equipment
should be budgeted at the project
level. Equipment should be
included in the project budget
to be charged to the project. The
total amount of equipment
should be included in the project
budget. The total amount of
equipment should be included
in the project budget. The total
amount of equipment should be
included in the project budget.

3.5 HUMAN RESOURCES DEVELOPMENT UNIT

The major thrust of the Human Resources Development Unit has been in the use of experiential learning. Structured experience for training in human behaviour has been extensively used for self improvement by trainees, both in the theory and in the organisational behaviour practicals. The concern in all the training programmes has been the understanding of human behaviour with a view to changing direct and control behaviour. Emphasis has also been laid on group behaviour in addition to individual behaviour. There has been a constant endeavour to bring about a general appreciation on the importance and usefulness of effective management of human resources, the most important resource of any organisation.

Objectives

- * to evolve effective and efficient approaches, methods and models in different areas of human resources development as interpersonal relationships, motivation, group dynamics, team-building, conflict management, leadership, organisational climate, institution-building and personnel management,
- * to develop information data base on various aspects of personnel management in the ICAR Institutes and SAUs for use in manpower planning, recruitment, training, performance appraisal and career development, and,
- * to undertake systematic review and study of management problems of ICAR Institutes and SAUs,

ACTIVITIES

Research Projects

1. " Culture of trust in research organisations"

(K. Pratap Reddy, M.M. Anwer, K.V.Subba Rao and Jagannadham Challa)

The data collection for all the 15 Institutes has been completed and statistical analysis and report writing is in progress. The report of the preliminary analysis of seven institutes was presented in the last annual report.

Courses/Programmes

Two Courses on Human Resources Development were conducted on August 22 to September 2, 1989 and December 12-23, 1989 for senior scientists of ICAR Institutes and Professors of Agricultural Universities, respectively. These programmes were coordinated by Dr. K.V. Subba Rao and Dr.M.M. Anwer.

A Five-day Management Development Programme for senior executives of ICAR in Personnel Management was conducted from February 13-17, 1990. The participants included Dy. Director-Generals, Asst. Director Generals of ICAR and Directors of ICAR Institutes. The programme was coordinated by Dr. K.V. Subba Rao.

3.6 INFORMATION AND DOCUMENTATION UNIT

Information and Documentation form the most important inputs in any form of scientific activity. Organised scientific information base is important for decision-making at all levels. Personal scientific information management and retrieval system is necessary for every research worker. The Information and Documentation Unit provides faculty support for almost all the training programmes of the Academy in addition to maintaining an excellent Library and Documentation Centre.

Objectives

- * to build a resource Library and Documentation Centre on agricultural research management and educational technology
- * to establish an information network of agricultural libraries in India
- * to set up a clearing house for user education/instruction, research on documentation and scientific information management
- * to assist in developing a policy framework for modernisation of libraries through research on and feedback from the present systems

Library

The Unit has established a well-organised specialised library. The resource library holds documents on Agricultural Research Management, Science Policy, Planning, Futurology, Forecasting Project Management, Educational Technology, Technology Systems and Policies, Women Studies, Research Communication, Information Management, Computer Technology, etc. International Corner and Foreign Tour Reports Section is a unique feature. The Library now holds about 20,000 documents and about 425 journals.

The Reprographic Section of the Unit is equipped with Plain Paper Copiers Scanning Machines and Microfilm Readers. Two more Plain Paper Copiers have been acquired during the year to meet the increased demand of reproduction facilities. About 2,25,000 photocopies from Plain Paper Copier and 1,500 faxils were cut from the scanning machines.

Computerisation

CDS/ISIS package version No 2.3 useful for the Information and Documentation activities developed by the UNESCO was procured and work on computerisation of Library holdings and other inhouse activities is in progress.

Faculty Support

Scientific Information Management is an integral part of most of the training programmes organised at the Academy. Information Management , Fact Finding Techniques, Review / Research Papers, Technical Writings, Frontiers of Information Services, Information Management tools for Curriculum Design and Development, Science Mapping, Research Front Specialities in the Sciences, Study Skills and other topics related to User Instructions were covered in this package.

ACTIVITIES

Workshops

A workshop on Information Technology and Agricultural Libraries Information Network was organised from December 4-8, 1989, for librarians of ICAR institutes and SAUs. Shri R.B. Gaddagimath coordinated the programme.

Research Projects

1. "Productivity of Indian Agricultural Research and Educational Institutes"

(R.B.Gaddagimath and T.Shanmuganathan)

Publications can be one of the tools to measure the scientific output of a research organisation. This project aims at studying the research output, analysing where they publish and identifying the significant journals. Lead institutions and their contributions to the overall agriculture development would be studied and analysed. Literature generated between 1975 and 1985 would be covered in this study. Later the CABI, AGRICOLA database would be covered and compared. It is planned to initiate further study for annual research indicators of all institutes under ICAR and State Agricultural Universities.

2. "Significant impact and core periodicals of interest amongst Indian Agricultural Scientists"

(R.B.Gaddagimath, Y.V.SivaPrasad and T.Shanmuganathan)

It is aimed to study, by survey, the popular significant and core journals of interest to the Agricultural Scientists. It is planned to bring out a discipline-wise significant journals in Agricultural Sciences. A handbook consisting of instructions for contributors would be brought out, as an outcome of this study.

3. "Directory of Indian Agricultural Library and Information Professionals"

(R.B.Gaddagimath, P.V.Nirmala and Y.Dhanalakshmi)

There has been no systematic attempt to compile a directory of Agricultural Library and Information professionals in India. In order to share the resources and also to identify the specialists, an exhaustive directory covering ICAR, SAUs and other national and international agricultural research/education organisations in India is proposed to be compiled.

Agricultural Research Management Abstracts (ARMA)

This is a unique half-yearly secondary publication brought out by the Academy broadly covering the areas related to agricultural systems and policies, project management, transfer of technology, educational technology, agricultural economics and information management. About 400 scientific and technical periodicals are scanned and significant papers are indexed and abstracted. The computer database on research management has been developed for an effective storage and retrieval for the benefit of research managers, educationalists and decision-makers. About 700 copies have been distributed for organisations involved in agricultural research, administration, management and policy-making. At present, the publication is distributed on exchange basis. Reprographic facilities for the original documents appearing in the abstracts are also offered at nominal cost. During the period under report, two issues were published and the remaining issue is in the process of being published.

FARM SERVICES UNIT

The campus of the Academy is being developed with an approach to functionality and aesthetics. The landscaping is done with a view to providing a congenial environment for academic activity. Stress is being laid in utilising a large and varied collection of trees, shrubs, climbers, herbaceous plants and ground covers with an eye on the educational value of a well labelled rich collection of flora.

The Agro-technology Demonstration Blocks have been conceived as a show window of the National Agricultural Research System. The latest technology and crop varieties released by different ICAR Institutes and State Agricultural Universities are put on show in one place for the benefit of trainees and visitors comprising national and international scientists and policy makers of agricultural research and development.

The Unit is also actively involved in establishing and demonstrating Non-conventional energy systems, particularly a domestic bio-gas plant, a windmill and a solar water heater.

Objectives

The Farm Services Unit has the objectives of landscaping the campus, developing and maintaining the estate and laying out the agro-technology demonstration blocks in different areas of crops and different silvi systems. The landscaping activities cover entire 50 ha. of the campus. The agro-technology demonstration blocks cover an area of 30 ha. out of which eight hectares are covered each by Field Crops and orchards and vegetable cultivation. The remaining area is covered by silvi systems, namely Agro-horticulture, silviculture, Horti-pasture, Agro-energy systems and plantations like High-density plantations, graded height plantations and green belts.

ACTIVITIES**Landscaping**

A lot of stress is given in making the campus green by way of planting and establishing lawns in and around the main gathering places like Administrative Block, Academic Block, Faculty Block and the residential areas of the participants, namely Halls of Residence and Scientists Home. Most of the

avenues on the major roads which include the tree species of about twenty genera are well established. A Tree Museum has been established with a collection of forest trees of about forty species.

Field Crops

The demonstrations in this particular area include Cereals, Millets, Pulses and Oilseeds. Sixteen varieties of different crops namely Maize, Jowar, Bajra, Redgram, Wheat, Sunflower and Groundnut were under demonstration. These include the currently recommended and latest varieties released by ICAR Institutes and Agricultural Universities.

Horticultural Crops

Apart from Guava, Sapota, Grape, Ber and Pomegranate, the five fruit crops established earlier, a mango orchard with a collection of 14 varieties, in an area of 0.50 Ha. which has accommodated 105 mango trees, was established.

The Guava orchard was expanded by including two more varieties, H8 and H9, released recently by IIHR. As a part of the demonstration of vegetable crops, seventy five varieties of twenty seven kinds of vegetables were grown.

Silvi Systems

Besides the conventional input systems for the increased productivity, several systems have been established in order to demonstrate the increase in productivity per unit area of land by way of obtaining food, fuel and fodder from the same piece of land. These include Agro-Horticultural, Horti-Pastoral System, Agro-Forestry and Silvi-Pastoral systems.

Further, a block of 100 mt x 50 mt has been planted with a recommended variety of Subabul T-8 at 2mt x 2mt spacing. This close plantation has been established to demonstrate the production of fodder and fuel in rural areas.

Graded height / Greenbelt Plantations

Graded height and Greenbelt plantations were established in peripheral and other areas to provide aesthetic environment to academic activity as well as for its educational value to the trainees in the campus.

PRODUCTION AND PUBLICATION UNIT

The Academy has taken up several publications on its own to help the research managers to keep abreast with the latest techniques of management. The Academy itself publishes the materials developed by the faculty for training purposes.

To facilitate the production and publication activities, the Academy is equipped with Reprographic Unit comprising Offset Printing Machine (Swift 300 Delux), Processing, Plate making and Binding systems. This Unit also has Screen Printing facility.

The Publications brought out at the Academy include :

1. Course Materials and Reading Materials for courses organised at the Academy
2. Case Materials prepared on the basis of research studies undertaken by the faculty

Besides the Academy brought out the following periodical publications:

ARMA (Agricultural Research Management Abstracts) is a half yearly publication appearing every June and December. It provides latest information in the specialised areas of Agricultural Research Systems and Policies , Project Management, Human Resources Management, Technological Development, Organisational Development, Technology Transfer, Agricultural Information , Agricultural Education and Agricultural Economics.

NAARM NEWS, published every quarter, brings out the information about the events programmes both present and future and welfare activities of the Academy.

NAARM ALUMNI NEWS, a half-yearly publication, brings out information for and about the alumni of the Academy.

GREEN AND GLORY, a laboratory journal published for every Foundation Course based on the theses of the trainees of a particular batch.

ADMINISTRATION AND ACCOUNTS

The Administrative Wing of the Academy is organised into the following six sections to provide effective administrative service:

1. Admn. I Section --- Establishment matters
2. Admn. II Section --- Cash and Bills
3. Admn. III Section --- Stores & Purchases, Transportation and Budget matters
4. Admn. IV Section --- Diary and Despatch
5. Admn. V Section --- All Training matters, compilation & publication of rules and regulations
6. Audit & Accounts --- All financial matters
Section

The Administrative Wing, besides ensuring smooth administration to help conduct the various training programmes of the Academy, also provides faculty support in the areas of Administration and Accounts in the training programmes.

3.8

THE HUMAN RESOURCES

The staff strength rose significantly during the year 1989-90. Three faculty members joined during this period. Similarly, eight members joined the other categories of staff.

The category-wise position of human resources during the period is set out in the following Table.

Name of the Post/ Category	Total Strength	Sanctioned as on 31.3.90	Posts filled
Scientific	43	21	
Technical	41	34	
Administrative	49	39	
Auxiliary	28	21	
Supporting	37	34	
Total	198	149	

3.9 FINANCIAL OUTLAY

Sl. No.	Head of Account	Actuals during 1988-89			Revised Estimates 1989-90		
		Non-Plan	Plan	Total	Non-Plan	Plan	Total
1.	Pay & Allowances	20,74,700	13,86,500	34,61,200	27,10,000	34,30,000	61,40,000
2.	Travelling Expenses	1,06,900	28,000	1,34,900	1,50,000	1,25,000	2,75,000
3.	Recurring Contingencies	21,68,800	9,07,800	30,76,600	21,00,000	2,50,000	23,50,000
4.	Non-Recurring Contingencies						
	A) Equipments	--	20,85,900	20,85,900	--	16,50,000	16,50,000
	B) Works	--	1,05,91,600	1,05,91,600	--	20,45,000	20,45,000
	C) Others	--					
	Total	43,50,400	1,49,99,800	1,93,50,200	49,60,000	75,00,000	1,24,60,000

4. UNDP ADVANCED CENTRE ON AGRICULTURAL EDUCATION AND RESEARCH MANAGEMENT

An Advanced Centre on Agricultural Education and Research was established at the Academy with the assistance of UNDP/UNESCO/FAO with a view to establishing a system for managing agricultural education and research. This Centre undertakes a number of activities to help make continuous adjustments to changing national needs through scientific study of educational research management process and training key personnel in their application.

The Centre conducted the following workshops since January 1, 1990

- * Effective Communication and Instructional Aids
- * Curriculum Design and Development
- * Effective Teaching-Learning Strategies for Undergraduate Veterinary Courses
- * Instructional Technology and Teaching Effectiveness
- * Methodologies and Approaches to Constraint Analysis in the Transfer of Agricultural Technologies
- * Information Technology and Agricultural Information Network

Dr. Shirley A. White, Professor, Cornell University, Ithaca, N.Y., U.S.A. was at the Academy from January 5 to February 11, 1989 and held discussions with the Director and the faculty regarding the training programmes being organised at the Academy. She also acted as the Facilitator of the workshop on "Effective Communication and Instructional Aids" conducted from January 31 to February 3, 1989.

Dr. Larry E. Miller, Professor of Agricultural Education, The Ohio University, Key Consultant to the Centre, was at the Academy from August 4 to September 5, 1989 and held discussions with the Director and Faculty members of the Academy. He was also Facilitator of the workshop on "Design and Analysis of Research Methods" held from August 28 to 31, 1989.

The UNDP/FAO Evaluation Mission, comprising Dr. Ermond Hartmans, UNDP Consultant and Dr. D.C. Kimmel, FAO Consultant, visited the Academy on March 7-8, 1990 and had discussions with the Director and

senior faculty members of the Academy. The Team also visited the various facilities created under the auspices of the Centre and reviewed its activities.

Under the UNDP Project on Advanced Centre for Post Graduate Agricultural Research and Education, seven faculty members completed advanced study in various Institutions and Universities in U.S.A .

The Academy has established the following laboratories to make the teaching-learning process more effective, and to help the trainees learn effective use of mass media for dissemination of information.

Closed Circuit Television Systems Laboratory

This laboratory provides intensive training to scientists in the preparation of effective resource materials for telecasting, conducting role play exercises for personality development and developing video lessons in subject-matter areas. The laboratory is also useful in developing proper speech communication.

Audio-Visual Lab and Photography Unit

The Unit, equipped with all modern photographic and audio reprographic facilities, provides an excellent opportunity to gain experience in the preparation and use of Audio-visual aids for effective presentation. These include Overhead Projectors, Epidiascope, Coloured Transparency-making units, Slide Projectors, Taperecorders and Public Address Systems.

Speech and Mass Communication Laboratories

These laboratories facilitate the scientists to record their programmes such as interviews, discussions and talks, specially for the rural listeners.

Organisational Behaviour Laboratories

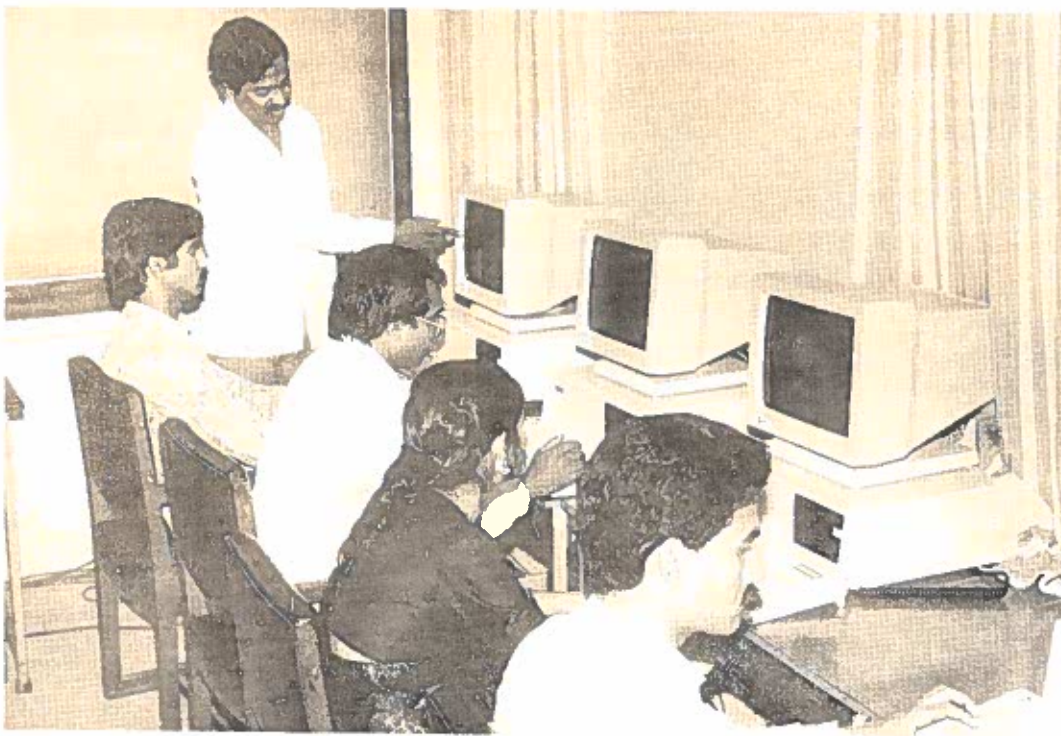
Individual and group exercises in team-building, leadership and group dynamics conducted in these laboratories provide an opportunity to trainees to understand the various aspects of organisational behaviour including human thought, action and behaviour.

Self-Instructional Carrels

These carrels, now being established at the Academy, provide an opportunity to the participants to undergo structured and programmed instruction under self-learning situations. A Telex Caramate imparting a tape slide programme, a VCP with monitor for a video tutorial and a computer for a programmed instruction are being integrated in each carrel for this purpose.



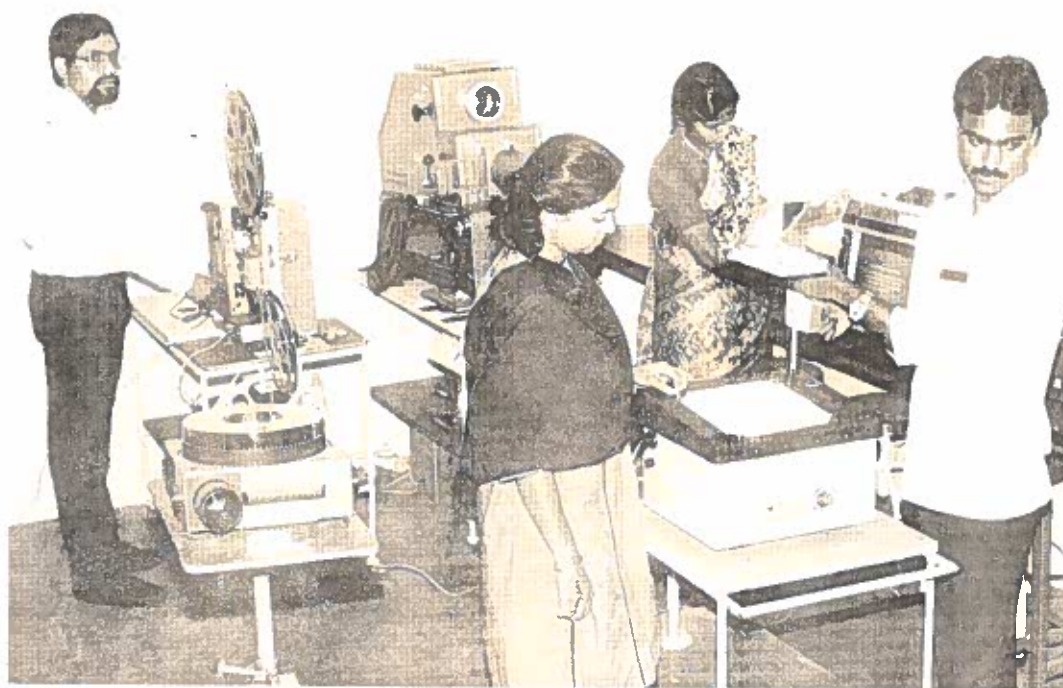
Photography Lab.



Computer Cell



Library



Audio Visual Lab

Software developed for learning Plant Pigments/Chlorophylls is being evaluated for its effectiveness as a learning tool.

Computer Cell

A Mini-frame Computer of WIPRO S-306 with 4 MB RAM and 170 MB Hard Drive along with 11 terminals having 640 K RAM has been installed in January 1990 in addition to the DCM TURBO 2000 PC-AT. Two dot matrix printers of FX series and a Near Letter Quality Printer were also acquired. These systems are used to provide intensive training on computers to the trainees of FCARPM, prepare Databases and develop Library Information Service. Besides, computers are also being used in Office Administration for preparation of Paybills, Accounting & Budgeting, Recruitment processing and secretarial work. The Faculty and Staff are also being trained on the use of various application packages such as Wordstar, dBASE and Lotus with a view to increasing the work efficiency.

Desk Top Publishing System

A Desk Top Publishing System, Apple Macintosh-SE has been installed recently. Several software packages including MS Word, Mc Write, Mc Paint and Mc Draw have also been acquired. These packages would help improve the quality of publications at the Academy.

Banking Facilities

The Central Bank of India, Golconda Fort Branch, Hyderabad, is operating an Extension Counter (Regular) at the Administrative Block to cater to the banking requirements of the Academy, Faculty members, staff and trainees. The counter is to be shifted shortly to the Faculty-cum-Laboratory building where safe deposit locker facility is proposed to be provided.

Capital Works

The construction of Faculty-cum-Laboratory building is fast nearing completion. A majority of the Faculty members have already shifted to the front block. The work on rear block is almost complete. Construction of residential quarters is also nearing completion. Work on 'Tractor/Vehicles' shed has been completed.

The total membership of the Association increased to 660 as on March 31, 1990. The General Secretary of the Association served as a member of the Steering Committee of CASAS-NAARM Asia Regional Seminar organised at the Academy from August 22 to 25, 1990. The Association also brought out two issues of 'NAARM Alumni News'.

Following are the office-bearers of the Association for 1988-90:

PATRON	:	Dr.N.S.Randhawa, Director General,
PRESIDENT	:	Dr.K.V.Raman, Director, NAARM
GENERAL SECRETARY	:	Shri R.Chittiraichelvan Scientist (SG)
VICE PRESIDENTS (2)		
Local	:	Dr P.B.Parthasarathy , Agricultural Economist A.P.A gricultural University Hyderabad.
All India	:	Dr U.K.Maheshwari, Scientist, Fresh Water Fish Farm of CIFE,Hoshangabad, Madhya Pradesh
JOINT SECRETARIES (5)		
Eastern Region	:	Dr. D.P. Singh Scientist Ramie Research Station J.A.R.I., Sorbhog, Assam

Northern Region	:	Dr. Rishendra Verma Scientist Indian Veterinary Research Instt. Izatnagar Uttar Pradesh
Western Region	:	Dr. B.P. Tiwari Ex-Director (Instruction) J.N.K.V.V. Jabalpur Madhya Pradesh
Southern Region	:	Sh. S. Damodaran, Scientist C.P.C.R.I. Kasargod, Kerala
Central Region	:	Sh. B.H. Singh Scientist I.A.S.R.I., Library Avenue New Delhi
Treasurer	:	Sh. C.P. Mohan Kumar Finance & Accounts Officer N.A.A.R.M. Hyderabad

7.1 Books

Samanta, R.K. (1989). Management in Agriculture and Rural Development, New Delhi, UDH Publishing House.

7.2 Scientific Papers

Raman K.V. and Balaguru T. (1989). Human Resources Development as a Key to the Success of FSR in India, Farming Systems Newsletter III (4) : 3-9.

Kishore Devesh. (1989). Curriculum Planning, Educational Technology Year Book, 1988. All India Association for Educational Technology, Nov.1989, New Delhi.

Kishore Devesh. (1989). Instructional Systems, Design & Methods. All India Association for Educational Technology, Kerala, Trivandrum.

Venkateswarlu K.(1989). Project Management Techniques for Agricultural Research and Developmental Projects. National Symposium On The Role of Scientific Research and Its Management in accelerating Socio-economic Transformation, Agricole Publishing Academy, P165-184.

Venkateswarlu K. (1989). Work Values of Agricultural Scientist. Decision, 16 (1)P 73-78.

Venkateswarlu K.(1989). Agricultural Scientists: What do they prefer in their job? Indian Journal of Applied Psychology. 26 (1). P73-78.

Samanta R.K. (1989). Scientists Perception of Motivational Climate in Agricultural Research Organisations vis-a-vis. Their job satisfaction and Role Perception - The Indian Scene, Indian Journal of Extension Education. 25(1 & 2).

Samanta R.K. (1989). Rural Development: Past Experience and Suggested Policy Frame, in Sequence of Development in North East India, J.B., Bhattacharjee (Ed), New Delhi, Omsons Publication.

Samanta R.K. (1989). People's participation in Rural Development. Khadi Gramodyog. 35(10).P 421-427

Samanta R.K. (1989). Repayment of Agricultural Credit, in Studies in Agricultural Extension and Management, G.L. Ray (Ed), New Delhi, Mittal Pub.

Samanta R.K. and R.C. Chelvan (1990). Development Communication -Key to Eradicate Rural Poverty. Khadi Gramodyog. 36(4). P 181-185.

Sabarathnam VE. (1988) Job Satisfaction and Efficiency of Dryfarm Scientist. Psychological Research Journal. 12(2) P 55-61.

Sabarathnam VE. (1989). An E.A.I Scale to Measure Job Satisfaction of Scientists. Indian Journal of Applied Psychology. 26 (1). P 48-60

7.3 Papers presented at Symposia / Seminars / Workshops / Conferences

Dr. K.V.Raman presented a paper on Human Resources Development as a Key to the Success of Farming Systems Research in India, at a Symposium on Farming Systems Research and Extension at the University of Arkansas, Fayetteville, Arkansas, from October 8-, 1989.

Dr.K.V. Raman and Dr.R Sudarsanam presented a paper on Curriculum Development for Higher Education in Agriculture: Arising Issues for 21st Century, at the Seminar on "Agricultural Education held at Haryana Agricultural University, Hissar from Feb.3-5, 1990.

Dr.K.V.Raman, Dr.T.Balaguru and Dr.R.K.Samanta presented a paper on Agricultural Research Systems for Future: The Indian context, Country Paper presented at CASAS-NAARM Asia Regional Seminar on Restructuring National Research Systems to meet Agricultural Development Challenges, held at NAARM, Hyderabad from August 22-25, 1989.

Dr.Jagdeesh C.Kalla presented a paper on Policy Intervention for the Development of Arid Zone Ecosystem, at the Seminar on Nehru and Economic Development held at the Rajasthan Agricultural University, Bikaner, November, 1989.

Dr. Devesh Kishore presented the following papers:

* "Model Approach on Curriculum Design and Development"at the Regional Seminar (Southern Region) on Educational Innovation for Development

organised at the State Council of Educational Research and Training, Tamil Nadu, Madras from March 27-31, 1989.

* "Planning & Education" at 14th Indian Social Science Congress held at New Delhi.

* "Research Management Process" at the workshop on "Women & Development : A Critical Appraisal of Household-level Research Methodologies" held at Baroda from July 5-14, 1989.

* "Development of Curriculum at macro and micro levels, Instructional Design and Methods, Distance Education, Distance Learning programme : Methodologies, and Case Studies" at the U.P. Academy of Administration from March 25-30, 1990.

Dr. K. Venkateswarlu presented the following seminar papers :

* "Project Management Techniques" at Agricultural Research Centre, Beltsville, Washington D.C. in January, 1989.

* "National Agricultural Research System in India: The role of NAARM" at the College of Agriculture, University of Illinois, U.S.A in April 1989.

* "Activities Slack Bar Chart" at the College of Engineering , University of Illinois, U.S.A. in April, 1989.

Dr. K.P.C. Rao presented a paper on Instructional Credit and Dryland Farmer - A Study in Gadwal Block of Mahaboobnagar District (Andhra Pradesh), at the 48th Annual Conference of Indian Society of Agricultural Economics at Banaras Hindu University, Varanasi in 1989.

Dr. Jagannadham Challa and Dr. T. Balaguru presented a paper on Management of Research and Development: A Case study from Agricultural Research, at the 77th session of the Indian Science Congress held at Cochin University of Science and Technology, Cochin from February. 4-9, 1990. Dr. Jagannadham Challa presented a paper on Changes in the site of Phosphorus Absorption in Calves at the 77th session of the Indian Science Congress held at Cochin University of Science and Technology, Cochin, from February 4-9, 1990.

8.

FACULTY DEVELOPMENT

8.1 Participation in Seminars / Workshops / Conferences

All the Faculty members participated in the CASAS-NAARM Asia Regional Seminar on "Restructuring National Research Systems to meet Agricultural Development Challenges" organised at the Academy from August 22-25, 1989.

The Director and all the Heads of Units participated in the First Academic Conference of the Association of Management Development Associations in South Asia (AMDISA) organised at the Administrative Staff College of India (ASCI) from February 26-29, 1990. The theme of the Conference was "Challenges of Management Development in South Asia".

Dr. V. Raghavendra Rao participated in workshop on "Women and Development: A Critical Appraisal of Household-level Research Methodologies" held at M.S. University, Baroda.

Dr. T. Balaguru participated in the following workshops:

- * National Workshop on Group Farming as a Basis for Efficient Resources Utilisation for Small and Marginal Farmers at the National Institute of Rural Development, Hyderabad on August 17-18, 1989.

- * International Workshop on Agricultural Research Management organised by ISNAR at The Hague, Netherlands, from November 5-10, 1989.

Dr. R.K. Samanta participated in the following seminars:

- * Special Seminar on 'Training needs of Managers and Management Development and Training of Trainers' organised by NIRD, Hyderabad, on May 5, 1989.

- * Seminar on "Perspective Issues and Experiences in Rural Development in North East India" organised by North East Indian Council for Social Science Research from May 26-27, 1989 at Shillong.

Dr. V.E. Sabarathnam attended the National Conference on Agricultural Education at Orlando, Florida, USA from November 30 to December 3, 1989.

Shri A.U. Gadewar participated in the following workshops/ seminars:

* Special Seminar on "Training needs of Managers and Management Development" organised by NIRD, Hyderabad on May 5, 1989.

* Regional Workshop on "Group Farming as a Basis for Efficient Resource Utilisation for Small and Marginal Farmers" held at National Institute of Rural Development, Hyderabad on May 25-26, 1989.

Dr. Jagannadham Challa, Assoc.Professor, participated in the Seminar on Organisational Dynamics for Rural Development held at NIRD, Hyderabad from Jan.3-6, 1990.

Shri R.B. Gaddagimath, Librarian, participated in the following Seminars/Symposia/ workshops:

* Fifth Conference of Association of College and Research Libraries (American Libraries Association), Cincinnati, Ohio, April 6-9, 1990.

* Society for Information Sciences National Conference on Local Area Network at Goa from January 18-20, 1990.

* Seminar on NTIS/Allied Technical Information Services at Hyderabad on February 6, 1990 as an observer.

8.2 Special Lectures Delivered

Dr. K.V.Raman, Director, delivered a special lecture on "Global Challenges to Food Production : Role of International Centres" at the Society of Scientists for Advancement of Agriculture, Hyderabad, on November 28, 1989.

Dr. Jagdeesh C. Kalla, Joint Director

1. 'Measuring benefits from watershed area development programmes' to the participants of the course on Agricultural Project Analysis, Planning and Review held at NIRD, Hyderabad on Oct.5, 1989.

2. 'Impact of agroecological zoning on planning processes in India' at NIRD, Hyderabad in March, 1990.

3. Sylvipastoral allocation of wastelands in India, to the participants of the Training Programme for Forest Development Officers held at NIRD, Hyderabad in October, 1989.

Dr. Devesh Kishore

1. 'How to plan and present lectures effectively?', at SVP National Police Academy, Hyderabad for police officers on Feb.7, 1989.
2. 'Farmers cooperation and their involvement in irrigated Agriculture at WALMTARI, Hyderabad on Feb.14, 1989.
3. 'Evaluation Techniques in the Service of Teaching learning continuum' at the Seminar conducted by Osmania University from March 11-13, 1989.
4. 'Problem Identification & Decision Making' and 'Preparation Formats for teaching learning process' at Osmania University, Hyderabad on March 13-14, 1989.
5. Keynote address on the inaugural day of Summer Institute at Rajendra Agril. University, Pusa (Samastipur), Bihar on May 10, 1989. Also delivered two lectures on the topics 'Case Studies' and 'Role Analysis' to the participants on 10th & 11th May, 1989.
6. 'Farm Broadcasting' and 'Media's Role in modernising agriculture' to the participants of the Training programme on Agril. Journalism and Public Relations at Extension Education Institute, A.P. Agril. University, Hyderabad on May 19th & 24th, 1989, respectively.
7. 'Curriculum Design and Development' at the Dept. of Family Resource Management, College of Home Science, APAU, Hyderabad on May 30, 1989.
8. 'Instructional Technology in Family Research Management' at College of Home Science, APAU, Hyderabad on June 5, 1989.
9. Information Technology: Choice of Options for Development Communication Practitioners to the participants of the workshop on 'Basic Communication and Reporting Skills, on June 19, 1989 at NIRD, Hyderabad.

10. "Instructional Technology the choice of options for development: communication practitioners' at NIRD, Hyderabad on June 21, 1989.
11. "Developmental Challenges: Need for shift in Training Programme on Developmental Perspectives" at NIRD, Hyderabad.
12. "Instructional Systems Design & Methods" at the Asian Regional Conference on "Educational Technology held at Trivandrum from Nov. 24-26, 1989.
13. "Development of Small and Marginal Farmers Strategies and Programmes" on Feb. 19, 1990 at NIRD, Hyderabad.

Dr. R.K. Samanta

1. 'Instructional aids for effective training and semantics in scientific communication' to the participants of short course on instructional Technology Use in Family Resource Management organised by Home Science College, APAU at Hyderabad on June 3, 1989.
2. 'Technology Transfer - A Systems Approach' to the participants of Rice Crop Management organised by DRR, Hyderabad on 26th August, 1989
3. 'Constraints in Technology Transfer' to the participants of the special course 'Socio-Economic Aspects of Irrigation Water Management' of WALAMTARI, Hyderabad on , January 24, 1990.
4. 'Audio-visuals in Extension and Communication' to the participants of 'National Training Course on Management of Extension Programmes' organised by MANAGE, Hyderabad on February, 17, 1990

Dr. V.E. Sabarathnam

1. 'A tested method for developing farm technology --an intersystem approach' at the University of Florida, Gainesville, U.S.A on November 29, 1989.
2. 'Role of Extension in Development of Agricultural Technology' at Iowa State University, Ames, Iowa, U.S.A on October 2, 1989.
3. 'Comparative analysis of Technology Development in developing countries and in USA' at Iowa State University, Ames, U.S.A on December 22, 1989.

Shri R.B. Gaddagimath

1. "Information Services on Dryland Farming, Agroforestry and Soils Management: Special reference to India" on Aug.26, 1989 at State Institute of Rural Development, Hyderabad.
2. "Publishing in the Third World India on the periphery of the International Knowledge Systems" at the Graduate School of Library and Information Science, University of Illinois, U.S.A. in April, 1989.
3. "Organisation of Agricultural Research and Education in India: The Role of NAARM" at International Agricultural Seminar at the INTERPAKS, University of Illinois, USA.
4. "Bibliographic instructions for the agricultural Scientists" at the GSLIS Staff Seminar, University of Illinois, USA on April 18, 1989.
5. "User studies and the user programmes for the scientists" to the students of Library and Information Science, Karnataka University, Dharwar.

** Dr. V.Raghavendra Rao received advanced training in U.S.A. from July 20, 1988 to Jan.12, 1989 in the field of Project Planning and prioritisation in the area of Agricultural Research Project Management under the UNDP Project on Advanced Centre for Post-Graduate Agricultural Research and Education. The training programmes he attended include: 'Project Implementation' for Agriculture and Rural Development; Auditing Courses on Project Management, Behavioural Factors--Process of Change and Conflict Management--Theory, Concepts and Methods.

Dr. Rao also visited USDA laboratories at Beltsville, Maryland to acquaint himself with Agricultural Research Service and programme implementation at area level and the related activities.

** Dr. T.Balaguru received advanced training in U.S.A. for a period of six months from August 3, 1989, in the field of Research Systems and Methodology in the area of Agricultural Research Project Management under the UNDP Project on Advanced Centre for Post Graduate Agricultural Research and Education.

Dr.Balaguru attended various programmes in the area of Agricultural Research Management in different American universities and Research Organisations. Some of the programmes attended by him include: Training of Trainers; Research Methodology in Management; workshop on Farming Systems Research Methodologies -- The Next decade; Special Programme on Agricultural Research Management and Extension Winter School. He also visited the Agricultural Research Service of the USDA at Beltsville, Maryland, U.S.A.

** Dr.K.Venkateswarlu received advanced training in U.S.A. for a period of six months from December 10, 1988 in the field of Network analysis in the area of Agricultural Research and Project Management under the UNDP Project on Advanced Centre for Post Graduate Agricultural Research and Education in U.S.A. He undertook Training Programme in Project Management Techniques (Network, PERT, CPM, etc.) at the University of Illinois. He had also undergone a course on Agricultural Research

Management in Washington D.C. at Beltsville Agricultural Research Centre for a period of one month.

**** Shri. R.B.Gaddagimath** received advanced training in U.S.A. for a period of five months from January 18, 1989 in the field of Information management systems under the UNDP Project. His major part of the training was at the University of Illinois, School of Library Science and did his internship at International Agricultural Library. The major emphasis of training was on Information Network, Library Automation and Resources Sharing through Information Technologies. He also visited Project LOEX - Eastern Mission University UMI - University Micro Film International; National Agricultural Library; National Medical Library; International Institute for Research; Agency for International Development (AID), Voluntary International Technical Assistance (VITA), and participated in the Fifth International Conference of Association of Colleges and Research Libraries) at Cincinnati.

**** Dr. VE.Sabarathnam** received advanced training in U.S.A. in the field of Technology Management in Agriculture for a period of four and half months under the UNDP Project at Iowa State University, Ames, Iowa, U.S.A. from July 31, 1989.

Dr. Sabarathnam specialised on the following aspects during the course:

Development Project implementation in International and Intercultural setting, Scientist and Development, Technology and Social Change, Technology Development, Diffusion Process in Agriculture, Indigenous Agricultural Knowledge, Sustainable Agriculture, Consequences of Farm Technology, Management of Job Stress and Improving Employees' Performance.

**** Dr. S.N.Saha** received advanced training in U.S.A. in the field of Agricultural Research Management for a period of four and half months under the UNDP Project on advanced centres from September 20, 1989 at the Ohio State University, Columbus, U.S.A.

He attended a number of courses including Introduction to Computer Science, Computers in Agricultural Decisions, Agricultural Policy and Farm Management., Seminar on Research: Application of Multivariate Statistical Analysis Techniques. He presented a seminar on Agricultural Research System in India. Dr. Saha visited the Agricultural Research Service and Economic Research Service of U.S. Department of Agriculture, Michigan State University, Research Foundation Centre of The Ohio State University, Columbus, Ohio Agricultural Research and Development Centre and The

Ohio State University Agricultural Technical Institute at Wooster, Ohio, to acquaint himself with the activities at these institutions.

**** Dr. P.Manikandan** received advanced training in U.S.A. for a period of four and half months from August 2, 1989 in the field of Agricultural Research Management.

Among the major training programmes he attended was the programme on "Training of Trainers for Agriculture and Rural Development", organised by the United States Department of Agriculture at George Mason University, Arlington, Virginia. Dr Manikandan also attended several courses related to Agricultural Research and Education Management at the University of Minnesota, Minneapolis.

He also interacted with the faculty and staff members at the University of Minnesota on various aspects related to Agricultural Research Management.

**** Dr. K.P.C. Rao** left for U.S.A. for advanced study in the field of Agricultural Research Systems with special reference to 'Assessment of returns to investments in Agricultural Research on February 3, 1990 under the UNDP Project on Advanced Centre for Postgraduate Agricultural Research and Education in U.S.A.

**** Dr.K.Vidyasagar Rao** attended one-year part time training programme "P.G.Diploma in Computer Applications" conducted by the Institute of Engineers, Hyderabad Chapter, Hyderabad from January 1, 1989 to January 1, 1990.

**** Dr. P.Kumar** attended the 15th Course in Management of Agriculture and Rural Development Programmes at Indian Institute of Management, Ahmedabad between December 8 to December 21, 1989.

** Dr. K.V. Raman, Director, NAARM, was a Visiting Senior Fellow at International Service for National Agricultural Research (ISNAR) for about three months from April, 1989. He was associated with all the activities of ISNAR as one of their staff and helped them in developing training materials for some of their programmes. He prepared a conceptual paper on " Systems Thinking in Planning Relevant Research" which has been brought out by ISNAR by staff Note No.89-62. He also brought out a paper on "Evaluation of Research Projects" based upon a Case Study on All India Coordinated Research Projects of the Indian Council of Agricultural Research.

** Dr. Raman also visited U.S.A. from October 8-11, 1989 to participate in a symposium on Farming Systems Research and Extension at the University of Arkansas, Fayetteville, U.S.A.

** Dr.T. Balaguru visited Netherlands from November 5-10, 1989, to participate in an international workshop on Agricultural Research Management organised by ISNAR at The Hague, Netherlands. He also held discussions with the Director General and senior faculty of ISNAR on matters related to organisation of Asia Regional Seminar on Management Information Systems to be held at the Academy in September, 1990.

Nehru Centenary Celebrations

Nehru Centenary Celebrations were organised at the Academy on June 28, 1989. Speaking on the occasion, Shri P.Vaman Rao, former Director of Information and Public Relations, Government of Andhra Pradesh, said Nehru's passion for scientific approach to economic problems and his emphasis on modern technology was instrumental in the rapid industrial progress achieved by India. Nehru was of the view that a responsive and dynamic society can emerge only on the inculcation of the scientific temper. The spectacular progress achieved by India in the application of science and technology is living testimony to the wisdom of Pt. Jawaharlal Nehru, he said.

Later, Shri Vaman Rao gave away prizes to the winners of Elocution and Essay Competitions on 'Nehru and the Scientific Temper'.

Presiding over the function, Dr. Jagdeesh C. Kalla, Director-in-charge, said that Nehru was an amalgam of traits which made him a true representative of his generation. Though Nehru was modern in his outlook by virtue of his western education, he was deeply conscious of India's rich heritage, he said.

Proposing a vote of thanks, Dr. Devesh Kishore, Chairman, NAARM Nehru Centenary Celebrations Committee, said 'Nehru was not a man but a procession of men. In him, he said, we witnessed a national hero, statesman, philosopher, historian, author and educator.

Hindi Day

The Academy celebrated Hindi Day on September 15, 1989 under the chairmanship of Dr. K.V. Raman, Director. Dr. Krishna Kumar Sharma, Chairman, Central Institute of Hindi, Hyderabad, was the chief guest of the day.

Dr. Raman, in his presidential address, exhorted the staff to rededicate themselves to the cause of greater implementation of Hindi in official work. He advised the staff to avail the various incentives offered by the Government to popularise Hindi. Dr. Raman gave away prizes to the winners of Hindi Essay writing competition organised on the occasion.

Dr.Sharma, Chief Guest, in his speech, said that we can love our National Language only when we love our country. He said care must be taken to ensure that the naturality and creativity in language is not lost in the process of translation. Hence, he advised, one should think in Hindi to express effectively in that language.

Dr.Devesh Kishore, Officer-in-charge, Hindi, explained the various activities of Hindi Unit.

The Academy was once again in limelight at the various Horticultural and Rose Shows conducted since January 1989 by bagging a number of trophies and prizes.

Horticultural Shows

Thirty seven prizes for the best collection of vegetable and fruit varieties, and cut flowers were won at the Horticultural Shows of 49th and 50th All India Industrial Exhibitions held in February 1989 and February 1990, respectively.

Rose Shows

The Academy bagged prestigious prizes such as Smt D.Indira Memorial Challenge Silver Trophy, Smt.T.Indira Subbarami Reddy Challenge Bowl, Coromandel Rolling Silver Trophy for the Best Fragrant Rose and the Sigma Cup for the Best Red Rose, and nineteen other prizes at the XIII Annual Rose Show organised by the Hyderabad Rose Society on January 8, 1989. The Academy also won the Best Red Rose Award and fourteen other prizes for various categories of exhibits in the Third Annual Rose Show conducted by the Horticulture Society, Secunderabad, on December 17, 1989. The Academy won the prestigious "FACT Shield" for the best Institutional Rose Garden and won eight other prizes for the different groups of roses and best display of roses in the Rose Show organised by the Hyderabad Rose Society on January 7, 1990.



Trophies and Awards won by the Academy at various Horticultural and Rose shows held in 1989 (top) and 1990 (bottom).



NEHRU AND THE SCIENTIFIC TEMPER
SPEAKER: Shri P.VAMAN RAO.
On June 28, 1989.



Dr. Jagdeesh C.Kalla, Jt. Director, Shri P. Vaman Rao, Chief Guest and Dr. Devesh Kishore at the Nehru Centenary celebrations.

राष्ट्रीय कृषि अनुसंधान प्रबंध अकादमी
हिन्दी दिवस समारोह
15 सितम्बर 1989



Dr K.K.Sharma, Chairman, Central Insititue of Hindi, Hyderabad (second from left) at the Hindi Day celebrations.



Dr. C.A. Akitoye, Commissioner for Agric. Coop, Lagos State, Nigeria, holding discussions with the Director.



Shri Hari Krishna Shastri, Union Minister of State for Agricultural Research and Education, planting a sapling at the Academic Block.



Shri Shyam Lal Yadav, Union Minister of State for Agriculture and Cooperation, seen with the Director.



The Director with the Sri Lankan delegation.

DISTINGUISHED VISITORS

** A Five member study team of trainers from Nepal visited the Academy on January 4, 1989. The team comprised Mr. S.K. Subedee, Mr. N.P. Regmi, Mr. R.P. Giori, Mr. B.B. Thapa and Mr. P.S. Prasad.

** Dr. Shirley A.White, Professor, Cornell University, Ithaca, N.Y.,U.S.A., visited the Academy from January 5 to February 11, 1989.

** Three senior officials of Afghanistan Government visited the Academy on January 11, 1989. The delegation included Mr. Abdul Aziz, President, Mr. Assadullah Bahar, Vice-President and Mr. Ahmad Zia, Director of Foreign Cooperation, Ministry of Agriculture, Jamal Mina, Kabul, Afghanistan.

** Mrs. R.P.K. Kannangara, Research Officer, Land and Water Use Division, Dept. of Agriculture, Peradeniya, Sri Lanka, visited the Academy on January 13, 1989.

** Dr. S.J.B.A. Jayasekera, Pulse Breeder, Agricultural Research Station, Pallekele, Kandasale, Sri Lanka, visited the Academy on January 13, 1989.

** Mr. Chhotubhai Patel, M.P. and Member of Governing Body of ICAR, visited the Academy on February 2, 1989.

** Mr. John Grant, Evaluation Officer, USAID, New Delhi, visited the Academy on February 22, 1989.

** A team of World Bank Representatives comprising Dr. R.V. Rama Krishna and Dr. P.C.S.Nair along with Dr. S.P. Ghosh, Project Director, NARP, visited the Academy on March 4, 1989, to review the participation of NAARM in National Agricultural Research Project.

** A two-member delegation from Sri Lanka visited the Academy on March 16, 1989. The members included Mr. M.M.N.D. Bandara, Project Director, Adzap, Anuradhapura, Sri Lanka and Mr. M.M. Premachandra, Assistant Director of Agriculture, Anuradhapura, Sri Lanka.

** Dr. N.Z. Milaschenko, Vice-Chairman, All Union Academy of Agril. Sciences (Moscow) VISHINIL; Dr. V.M. Penchukoo, All Union Research Institute of Oilseeds Crops and Dr. P.V. Solomakhin, Asst. Agril. Attache, USSR Embassy, New Delhi, visited the Academy on April 19, 1989.

**** A Soviet Team of Dr. V.Afanasyev, Dept. of USSR GOSAGROPROM, Dr. P. Sorokin, Dean Moscow College of Agricultural Engineering, Moscow,USSR and Dr. M. Ilyukhin,; Moscow Institute of Agril. Engineers visited the Academy from May 13-15, 1989 to study methods of personnel training in Management of Scientific Research.**

**** Shri Hari Krishna Shastri, Union Minister of State for Agricultural Research and Education, visited the Academy on May 22, 1989. Dr Jagdeesh C.Kalla, Director-in-charge,introduced the Faculty members and apprised him of the activities at the Academy.**

**** A World Bank team visited the Academy on May 15, 1989. The team included Dr. Mc Ginis, Consultant, General Crop Research, Dr. K.L. Chadha, Dy. Director-General (Hort.) Coordinator. Dr. J.P Tandon, Project Director (Wheat) and Dr. T.C. Jain of World Bank.**

**** Dr. Karim B. Kolawole, Dy. Senior Agril. Res. and Extension Coordinator and Mr. J.A. Oshakuade, Dy. Agril. Extension Coordinator from Federal Agricultural Coordinating Unit, Nigeria, visited the Academy on June 30th, 1989.**

**** Dr.Larry E. Miller, Professor of Agricultural Education, The Ohio University, U.S.A. and Key Consultant to the Advanced Centre, on Agricultural Education and Research , was at the Academy from August 4 to September 5, 1989.**

**** Five member delegation from Bangla Krishi Bank visited the Academy on August 5, 1989. The delegation comprised 1. Shri G.M. Salahuddin Ahmed, Asst. General Manager, Planning Research and Evaluation Department, Bangladesh Krishi Bank; 2. Shri Qazi Aziz Ahmed, Asst. General Manager, Regional Office, Comilla, Bangladesh Krishi Bank; 3. Shri M.Fareduddin, Asst. General Manager, Agrabad, Chittagong; 4. Shri Said Ahmed, Regional Manager and 5. Shri Sayed Hossain, Executive Engineer, Hobhad.**

**** Dr. Stephen D. Biggs of the Overseas Development Group from the University of East Anglia, Norwich, UK, visited the Academy from August 8-12, 1989 and discussed with the faculty members on the role of Management Information Systems in Agricultural Research.**

**** Shri Shyam Lal Yadav, Minister of State for Agriculture and Cooperation, Govt. of India, New Delhi,visited the Academy on August 14, 1989. Dr**

K.V.Raman, Director, accorded a warm welcome to the Minister and explained the various activities of NAARM.

** Several distinguished scientists from home and abroad visited the Academy to participate in the CASAS-NAARM Asia Regional Seminar on "Restructuring National Research Systems to Meet Agricultural Development Challenges" from August 22-25, 1989. They include:

Dr. Abdul Halim Hassan, Dy. Director General, Palm oil Research Institute of Malaysia, Kuala Lumpur, Malaysia

Dr. Howard Elliot, Director General, ISNAR, The Hague, Netherlands

Dr. Samion Abdullah, MARDI, Kuala Lumpur, Malaysia

Dr. Md. Yusof Shahbuddin, Rubber Research Institute of Malaysia, Kuala Lumpur, Malaysia

Dr. Beng Paik, Dept. of Agriculture, Kuala Lumpur, Malaysia

Dr. Syed Anwarul Haque, Bangladesh Agril. University, Mymensingh, Bangladesh

Dr. Md. Shahidul Islam, Bangladesh Agril. Research Institute, Gazipur, Bangladesh

Mr. Lee Song Kheun, Agriculture Division, Singapore

Dr. Md. Yussof Bin Haji Mohiddin, Dept. of Agriculture, Brunei Darussalam

Dr. R.P. Singh, Pro-Vice Chancellor, N.D. University of Agriculture and Technology, Faizabad, U.P.

Dr. A. Ahmed, Vice Chancellor, S.K. University of Agril. Science and Technology, Jammu & Kashmir

** A four member study group on 'Legumes and Coarse grains' from Myanmar Agricultural Service, Myanmar (Burma) comprising Dr. Tun Saing; Dr. K. Lay and Dr. T.T. Alaing and O. Than visited the Academy on Sept. 2, 1989 and held discussions with faculty.

** Three member Nigerian delegation comprising Dr. Godwell Ogboghodu, Hon. Commissioner for Education, Mr Michael Olotu, Head of Agril. Development Projects and Mr Gregory Onaghise, Project Manager, Agril. Development Projects visited the Academy on September 16, 1989.

** Dr. Oumar Tall, Director-General, I.E.R.D. Bamath - Mali B.P.258 (West Africa) visited the Academy on Sept. 28, 1989.

** Dr. De La Cruz, UNESCO, Bangkok, Thailand, visited the Academy on Sept. 29, 1989.

** Dr. Roger Slade, World Bank, visited the Academy on November 16, 1989.

** William B. Feidt, National Agricultural Lib., Beltsville, USA, visited the Academy on December 8, 1989.

** Dr. George H. Axinn, FAO Representative, New Delhi visited the Academy on December 28, 1989.

** Dr. C. A. Akitoye, Commissioner for Agric. Coop, Lagos state, Nigeria, visited the Academy on Jan. 19, 1990.

** Dr. John R. Myers of the Current Research Information System (CRIS) from the USDA, Washington D.C., USA visited the Academy on January 23-24, 1990 and held discussions with the faculty members on the research management information need assessment of the ICAR including the possible role of CRIS in helping ICAR to meet those needs.

** Dr. Byron T. Mook, Senior Research Officer, ISNAR, The Hague, Holland visited the Academy on February 9, 1990

** Dr. Geoffrey Swenson, Consultant, ISNAR, visited the Academy on February 9, 1990.

** Dr. Ermond H. Hartmans, UNDP Consultant and Dr. D.C. Kimmel, FAO Consultant visited the Academy on March 8, 1990.

** Dr. Richard P. Masilomoney, Vice Chancellor, Veterinary and Animal Sciences, University of Tamil Nadu and Dr. M.N. Menon, Chairman, Expert Committee, TNVASU, visited the Academy on March 28, 1990.

** Dr. C.K. Rao, Member of Expert Committee and Former Vice Chancellor of Punjab Agricultural University, visited the Academy on March 28, 1990.

**Steering Committee on Advanced Centre on
Agricultural Education and Research Management.**



**From L to R : Dr. V.Raghavendra Rao, Dr.K.V.Subba Rao,
Dr. Jagdeesh C.Kalla, Dr. R.Nagarcenkar, Dr.K.V.Raman,
Shri V.Kumar, Dr.Devesh Kishore, Shri K.L.Aneja**

Management Committee



**From L to R : Shri K.L.Aneja , Dr.Jagdeesh C.Kalla, Shri N.B.Reddy,
Dr. N.K.Bhattacharya, Dr.K.V.Raman, Dr.D.Bap Reddy, Dr.A Appa Rao,
Dr.V.R.P.Sinha.**

Management Committee

The Academy is guided in planning, organising and conducting of various types of training programmes and research studies by the Management Committee of NAARM. The Management Committee approves the programmes to be undertaken. The Academy ensures their implementation both in terms of physical targets and schedules and provide broad guidelines for future programmes. The Management Committee also helps in formulation and consideration of proposals for Five Year Plan and Annual Plan and takes up periodical review of progress of development schemes.

The Management Committee, besides assisting Director of the the Academy in the discharge of his functions, concentrates his attention on the Research and other programmes of the Academy and ensures their implementation. While examining these programmes periodically, it pinpoints bottlenecks, if any, and suggests suitable remedial measures.

The 13th, 14th and 15th meetings of the Management Committee of the Academy were held on June 25, 1988; January 28, 1989 and September 30, 1989, respectively. The composition of the Management Committee of NAARM is as follows:

- | | |
|---|----------|
| 1. Dr.K.V.Raman
Director, NAARM
Rajendranagar
HYDERABAD 500 030 | Chairman |
| 2. Dr. Jagdeesh C. Kalla
Joint Director
NAARM, Rajendranagar
HYDERABAD 500 030 | Member |
| 3. Dr.N.K.Bhattacharya
Director
Central Institute for
Research on Goats
P.O. FARAH 281 122
Mathura (U.P) | Member |

- | | |
|--|--------|
| 4. Dr.V.R.P.Sinha
Director
Central Institute of
Fisheries Education
Versova, BOMBAY 400 058 | Member |
| 5. Dr.N.T.Singh
Director
Central Soil Salinity
Research Institute
KARNAL 132 001 (Haryana) | Member |
| 6. Shri P.Ramakanth Reddy, IAS
The Commissioner & Director of
Agriculture
HYDERABAD | Member |
| 7. The Director of Animal Husbandry
Govt. of Uttar Pradesh
LUCKNOW (U.P) | Member |
| 8. Dr.A.Appa Rao
Vice-Chancellor
A.P.Agricultural University
Rajendranagar
HYDERABAD 500 030 | Member |
| 9 Shri N.B.Reddy
Post : Aurad
(Sirsi), Dist. Bidar
Karnataka | Member |
| 10. Dr.D.Bap Reddy
33/4 RT, Barkatpura
HYDERABAD 500 027 | Member |
| 11. The Deputy Director General (Edn)
Indian Council of Agril. Research
Krishi Bhavan
NEW DELHI 110 001 | Member |

12. Shri S.W.Oak
Director (Finance)
Indian Council of Agri. Research
Krishi Bhavan
NEW DELHI 110 001

Member

13. Shri K.L.Aneja
Chief Admn. Officer
NAARM, Rajendranagar
HYDERABAD 500 030

Member-Secretary

Steering Committee on Advanced Centre on Agricultural Education and Research Management

This Committee was initially called the Sub-Project Advisory Committee and has been later renamed as Steering Committee. The 4th and 5th meeting of the Steering Committee on the Advanced Centre on Agricultural Education and Research Management were held on February 24, 1989 and December 4, 1989, respectively, at NAARM, Hyderabad. The committee reviewed the progress of the project and provided broad guidelines for its implementation based on the suggestions of user agencies. The following are the members:

1. Dr.K.V.Raman
Director
National Academy of Agril.
Research Management
Rajendranagar
HYDERABAD 500 030

Chairman

2. Dr.Maharaj Singh
Deputy Director General (Edn.)
Indian Council of Agril. Research
Krishi Anusandhan Bhavan
NEW DELHI 110 012

Member

3. Dr.N.K.Anant Rao
Retd. DDG(Edn.), ICAR
C/o Dr.R.J.Galagali
Plot No.84, Sector 12
M.M.Extension
BELGAUM 590 016

Member

- | | |
|--|--------|
| 4. Dr.A.Appa Rao
Vice-Chancellor
A.P.Agricultural University
Rajendranagar
HYDERABAD 500 030 | Member |
| 5. Dr. A.M. Michael
Director
Indian Agril. Research Institute
NEW DELHI 110 012 | Member |
| 6. Dr.R.Nagarcenkar
Director
National Dairy Research Institute
KARNAL 132 001 | Member |
| 7. Dr.A.Ram Mohan Rao
Dean, P.G.Studies
A.P.Agricultural University
Rajendranagar
HYDERABAD 500 030 | Member |
| 8. Dr.K.V.Subba Rao
NAARM
Rajendranagar
HYDERABAD 500 030 | Member |
| 9. Dr.Devesh Kishore
NAARM
Rajendranagar
HYDERABAD 500 030 | Member |
| 10. Dr.V.Raghavendra Rao
NAARM
Rajendranagar
HYDERABAD 500 030 | Member |
| 11. Dr.Guy Baird
Indian Coordinator
WINROCK International
Support Services Contract
Ashok Hotel, Room No.1536
Chanakyapuri
NEW DELHI 110 021 | Member |

12. Sh.Md.Habeebuddin
Asst. Admn. Officer
NAARM, Rajendranagar
HYDERABAD 500 030

Member

13. Shri K.L.Aneja
Chief Admn. Officer
NAARM, Rajendranagar
HYDERABAD 500 030

Member-Secretary

The Steering Committee of NAARM, was reconstituted for a period of three years with effect from 01.01.90 with the following as its members:

- | | |
|---|---------------------|
| 1. Dr.K.V.Raman, Director, NAARM, Hyderabad. | . Chairman |
| 2. Dr.P.B.Mathur, Dy. Director General (Edn.) | .. Member |
| 3. Dr.A.Appa Rao, Vice-Chancellor, APAU | .. Member |
| 4. Dr.U.C.Upadhyay, Vice-Chancellor
Narendra Dev University of Agriculture
and Technology, Faizabad | .. Member |
| 5. Dr.S.Jayaraj, Vice Chancellor, Tamil Nadu
Agricultural University, Coimbatore | .. Member |
| 6. Dr.R.C.Mehta, Dean (Agriculture)
Rajasthan College of Agriculture (RAU)
Udaipur | .. Member |
| 7. Dr.O.N.Kunzru, Jt. Director, Indian
Veterinary Research Institute, Izatnagar, U.P. | .. Member |
| 8. Dr.P.Geervani, Dean of Home Science
A.P.A.U., Rajendranagar, Hyderabad | .. Member |
| 9. Dr.Maharaj Singh, Winrock International
P.O.8809, 7 Poorvi Marg, Vasant Vihar
New Delhi | .. Member |
| 10. Dr.Jagdeesh C. Kalla, Joint Director
NAARM, Rajendranagar, Hyderabad 500 030 | .. Member |
| 11. Dr.Devesh Kishore, Professor, NAARM
Rajendranagar, Hyderabad 500 030 | .. Member |
| 12. Dr.V.Raghavendra Rao, Professor, NAARM
Rajendranagar, Hyderabad 500 030 | .. Member |
| 13. Dr.George H. Axinn, FAO Representative
in India & Bhutan, FAO of the United
Nations, 55 Max Mueller Marg (ex-Lodi
Estate), New Delhi | .. Member |
| 14. Shri K.L.Aneja, Chief Admn. Officer
NAARM, Rajendranagar, Hyderabad 500 030 | .. Member-Secretary |

Quinquennial Review Team

The Quinquennial Review Team reviews the performance of the Academy with regard to achievement of its objectives in the last five years. It also examines and makes recommendations on the objectives, scope and relevance of the research programmes for the next five years in relation to the overall national plans, policies and long and short-term priorities. The team comprising the following members visited the Academy from January 2-6, 1989:

- | | |
|--|----------------------|
| 1. Dr. D.P. Singh, former Vice-Chancellor,
Pantnagar and Bihar Agricultural
Universities | ... Chairman |
| 2. Dr. V. Rajagopalan, former Vice-Chancellor,
TNAU, Coimbatore | Member |
| 3. Dr. Vinayshil Gautam, Professor, Centre
of Management Studies, IIT, Delhi | Member |
| 4. Dr. Ramamurthy, former Vice-Chancellor,
OUAT, Bhubaneswar | Member |
| 5. Dr. Shankar Narayan, former Vice-Chancellor,
Bangalore University & Hony. Professor,
IIM, Bangalore | ... Member |
| 6. Dr. Maharaj Singh, DDG (Edn.), ICAR | ... Member-Secretary |

The Team held discussions with the Faculty of the Academy as well as academicians and experts of the local institutes who have been associated with the NAARM, regarding training programmes.

Consequent on superannuation of Dr. Maharaj Singh in February, 1989, Dr. Jagdeesh C. Kalla, Jt. Director, NAARM, was appointed as Member-Secretary.

Official Language Implementation Committee

The Official Language Implementation Committee of the Academy provided periodical suggestions for implementation of Official Language policy and rules and regulations of Govt. of India that are issued from time to time. The following are the members:

- | | |
|--|-------------|
| 1. Dr. Devesh Kishore
Principal Scientist | .. Chairman |
|--|-------------|

- | | | |
|--|----|------------------|
| 2. Dr. K.V.Subba Rao
Principal Scientist | .. | Member |
| 3. Shri K.L. Aneja
Chief Administrative Officer | .. | Member |
| 4. Dr. P. Manikandan
Scientist Selection Grade | .. | Member |
| 5. Shri D. Venkateswarlu
Jr. Hindi Translator | .. | Member-Secretary |

Institute Joint Staff Council

The main objective of the Institute Joint Staff Council meeting is to promote harmonious relations between the Institutes of ICAR and its employees with a view to increasing efficiency of the service.

Following are the members of the Council:

Dr.K.V.Raman, Director	... Chairman
------------------------	--------------

OFFICIAL SIDE

- | | |
|---|---------------|
| 1. Shri K.L.Aneja, Chief Administrative Officer | ... Secretary |
| 2. Shri C.P.Mohan Kumar, Finance and Accounts Officer | ... Member |
| 3. Dr. Devesh Kishore, Principal Scientist | ... Member |

STAFF SIDE

- | | |
|----------------------------|---------------|
| 1. Dr.K.Venkateswarlu | ... Member |
| 2. Shri N.R.Nageswara Rao | ... Member |
| 3. Smt. P.V.Nirmala | ... Member |
| 4. Shri B.Ch Satyanarayana | ... Member |
| 5. Shri.P.Swamy | ... Member |
| 6. Shri.M.Narasinga Rao | ... Member |
| 7. Shri P.P.Brahmaji | ... Secretary |

NAARM Grievance Committee

The Grievance Committee serves as a forum for employees to ventilate their personal grievances relating to official matters.

Following are the members of the NAARM Grievance Committee:

- | | |
|---|--------------|
| 1. Dr.Jagdeesh C.Kalla,
Jt.Director | ...Chairman |
| 2. Dr.K.V.Subba Rao,
Head, Human Resources
Development Unit | ...Member |
| 3. Shri. K.L. Aneja
Chief Admn. Officer | ...Member |
| 4. Shri. C.P. Mohan Kumar
Finance & Accounts Officer | ...Member |
| 5. Shri. R.C. Chelvan
representing
Scientific Category | ...Member |
| 6. Shri. N.R. Nageswara Rao
representing
Technical Category | ...Member |
| 7. Shri. S. Swamy
representing
Auxiliary Category | ...Member |
| 8. Shri. K.S. Sharma
representing
Administrative Category | ...Member |
| 9. Shri. B.H. Dharm Raj
representing
Supporting Category | ...Member |
| 10. Asst.Admn. Officer | ...Secretary |

K.V.Raman

Director

Director's Personal CellL.Jhansi Lakshmi
T.SrinivasSr.Stenographer
Jr.Clerk**AGRICULTURAL RESEARCH SYSTEMS AND POLICIES UNIT**Jagdeesh C.Kalla
P.Kumar
VE.Sabarathnam
T.Balaguru
K.P.C.Rao
K.Vidyasagar Rao
M.J.Pradeep Kumar
K.Venkateswara Kumar
C.V.V.S.S.R.L.Phani Raj
Girija KumariJoint Director & Unit in-charge
Principal Scientist
Scientist SG
Scientist SG
Scientist SG
Scientist SG
Computer Operator
Statistical Assistant
Jr.Clerk
Jr.Stenographer (From 03.10.89)**EDUCATIONAL SYSTEMS AND TECHNOLOGY UNIT**Devesh Kishore
R.Sudarsanam
K.M.Reddy
A.Gopalam
N.R.Nageswara Rao
Ch.Janardhan Rao
N.Naresh KumarPrincipal Scientist & Unit in-charge
Principal Scientist (From 05.07.89)
Scientist SG
Scientist SG (From 15.07.89)
A.V.Operator
Video Cameraman (From 07.11.89)
VCR Telecine Operator
(From 30.09.89)M.Venkatesh
K.Ajay KumarJr. Stenographer
Technician (Instruments)**TRANSFER OF TECHNOLOGY SYSTEMS
AND POLICIES UNIT**R.K.Samanta
R.ChittiraichelvanScientist SG & Unit in Charge
Scientist SG

A.U.Gadewar
L.Venkateswarlu
D.Rajagopal Rao
Bansidhar Nayak
A.Vijayalakshmy

Scientist SG
Photographer
Media Operator
Projectionist
Jr.Stenographer

PROJECT MANAGEMENT UNIT

V.Raghavendra Rao
S.N.Saha
K.Venkateswarlu
P.Manikandan
A.Mercy

Principal Scientist & Unit in-charge
Principal Scientist
Scientist SG
Scientist SG
Jr.Stenographer (From 21.09.89)

HUMAN RESOURCES DEVELOPMENT UNIT

K.V.Subba Rao
M.M.Anwer
Jagannadham Challa
Sarada Samanta

Principal Scientist & Unit in-charge
Scientist SG
Scientist SG (From 30.09.89)
Sr.Stenographer

INFORMATION AND DOCUMENTATION UNIT

R.B.Gaddagimath
P.V.Nirmala
Y.V.Siva Prasad
T.Shanmughanathan
P.Ananda Kumar
Y.Dhanalakshmi
Venkatesham

Librarian, & Unit in-charge
Cataloguer
Documentation Assistant
Library Assistant
Jr.Stenographer (From 11-10-89)
Jr. Clerk
Xerox Machine Operator

CENTRALISED SERVICES

Farm Services

V.Murali
Mohd. Abdul Basith
K.Rangaswamy
B.Veeraiah
A. Laxman Maharu

Farm Manager
Lab Technician (On study leave)
Field Assistant
Farm Assistant
Farm Assistant

K. Prabhudas
K. Thirumalaiah
M. Padmaiah
Veeranarasiah
P. Eswari

Jr. Clerk
Pump House Operator
Tractor Driver
Carpenter cum Painter
Power Tiller Driver

PRODUCTION AND PUBLICATIONS

R. V. V. S. Prakash Rao
N. Narasimha Reddy
A. Murali Mohan Rao
K. Anil Kumar
M. Sekhar Reddy
S. Swamy
Prashant Gaikwad
C. Chandramouli

Asst. Editor
Offset Machine Operator
Artist
Reprographic Assistant
Dark Room Assistant
Assistant Gestetner Operator
Binder
SSG-1

ADMINISTRATION AND ACCOUNTS

K. L. Anuja
C. P. Mohan Kumar
Mohd. Habeebuddin
M. Srinath Reddy
B. Ch. Satyanarayana
M. V. V. S. Ravindra Babu
K. Janaki Ramaiah
G. Narasimha Reddy
D. Venkateswarlu
T. Ashok Kumar
Y. Shankar Rao
K. S. Sarma
P. P. Brahmaji
K. Satya Priya
A. P. Ramachandran
J. Renuka
C. Bagaiah
G. Kohad
W. Srinivas Bhat
M. Narasimha Rao
K. Swarajya Lakshmi
N. Raghunath
M. Dinesh

Chief Administrative Officer
Finance and Accounts Officer
Asst. Admn. Officer
Public Relations Officer
Security Officer-cum-Caretaker
Electrical Overseer
Superintendent
Superintendent
Jr. Hindi Translator
Assistant
Assistant
Assistant
Assistant
Assistant (Upto 10.11.89)
Hindi Asst. (From 21.12.89)
Sr. Clerk P.
Sr. Clerk
Sr. Clerk
Sr. Clerk
Jr. Library Assistant
Jr. Steno
Jr. Clerk

P. Srinivas
 K. Srinivasa Rao
 T. Pardhasarathy
 G. Raj Reddy
 B. Padma Saroja
 S. Balakamesh
 P. Rajarajeshwari Jyothi
 K. Surekha
 V. Govinda Raju
 P.D.Jessie Ecclcia
 L. Ramesh
 K. Obulpathi
 M. Mohan Rao
 M.K.Samshuddin
 B. Satyanarayana Murthy
 Y. Venkateswara Rao
 M. Sonkusare
 A. Nageswara Rao
 N. Prabhakar Rao
 K. Sattiah
 S.K. Peera
 Mohd. Naseer Khan
 G. Muthyalu
 U. Venkataratnam
 N. Ashok
 M.M.Mohan Rao
 Ramiah Sahadev Rao

Jr. Clerk
 Jr. Clerk
 Jr. Clerk
 Jr. Clerk
 Jr. Clerk
 Jr. Clerk
 Jr. Clerk
 Jr. Clerk
 Jr. Clerk
 PBX Operator(From 13.02.90)
 Electrician cum Plumber
 Technician, Electrical
 Technician, Electrical
 Technician, Electrical
 Technician, Electrical
 Technician, Electrical
 Technician, Electrical
 Lineman
 Plumber
 Driver
 Driver
 Driver
 Driver
 Driver
 Driver
 Driver
 Vehicle Mechanic
 Pump Driver

Hostel Services

Zameer Ahmed
 Sham Bahadur
 P. Neelakantam
 D. Sangamma
 B. Subba Rao
 B. Santamma
 K. Kalavathi
 S. Sakuntala
 G. V. Bikshapathi
 Phool Kumar
 A. Yadaiah
 M. Krishnaiah

Manager(Hostel Services)
 Catering in-charge
 Sr. Clerk
 Cook SSG(II)
 Asst. Cook (SSG I)
 SSG I
 SSG I
 SSG I
 SSG I
 SSG I
 SSG I
 SSG I

G. Dasaradha	SSG I
C. Anjaiah	SSG I
T. Jangamma	SSG I

Other Supporting Services

G. Mani Bai	SSG II
B. Narayana	SSG II
B. H. Dharmaraj	SSG II
P. Swamy	SSG I
R. Sattaiah	SSG I
P. Bal Raj	SSG I
Khalid	SSG I
M. Yadaiah	SSG I
K. Satyanarayana	SSG I
M. Ashok	SSG I
N. Sugunamma	SSG I
C. Bikshapathi	SSG I
M. Narasing Rao	SSG I
S. Jangaiah	SSG I
B. Bharatamma	SSG I
J. Chandraiah	SSG I
G. Pentaiah	SSG I
M. Yadamma	SSG I
M. Sham Rao	SSG I
M. Eswaramma	SSG I
L. Satyanarayana	SSG I
I. Bharatamma	SSG I

